

1 Nicheformer: a foundation model for single-cell and spatial omics

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17 Abstract

18 Tissue makeup and the corresponding orchestration of vital biological activities, ranging
19 from development and differentiation to immune response and regeneration, rely
20 fundamentally on the cellular microenvironment and the interactions between cells. Spatial
21 single-cell genomics allows probing such interactions in an unbiased and, increasingly,
22 scalable fashion. To learn a unified cell representation that accounts for local dependencies
23 in the cellular microenvironment and the underlying cell interactions, we propose to
24 generalize recent foundation modeling approaches for disassociated single-cell
25 transcriptomics to the spatial omics setting.

26 Our model, Nicheformer, is a transformer-based foundation model that combines human
27 and mouse dissociated single-cell and targeted spatial transcriptomics data to learn a
28 cellular representation useful for a large variety of downstream tasks. Nicheformer is
29 pretrained on over 57 million dissociated and 53 million spatially resolved cells across 73
30 tissues from both human and mouse. Subsequently, the model is fine-tuned on spatial
31 tasks for spatial omics data to decode spatially resolved cellular information.

32 We demonstrate the usefulness of Nicheformer in both zero-shot-like as well as fine-tuning
33 scenarios on a novel set of spatially-relevant downstream tasks such as spatial density
34 prediction or niche and region label prediction. In particular, we show that Nicheformer
35 enables the prediction of the spatial context of dissociated cells, allowing the transfer of
36 rich spatial information to scRNA-seq datasets. We define a series of novel spatial
37 prediction problems and observe consistent top performance of Nicheformer,
38 demonstrating the advantage of the improved model capacity of the underlying transformer.
39 Altogether, our large-scale resource of more than 110 million cells in a partial spatial
40 context, together with the set of novel spatial learning tasks and the Nicheformer model
41 itself, will pave the way for the next generation of machine-learning models for spatial
42 single-cell analysis.

44 Introduction

45 Single-cell genomics technologies have advanced our understanding of cellular heterogeneity of entire
46 tissues, organs, and organisms. Large-scale data generation efforts have led researchers to chart atlases of
47 cellular heterogeneity of specific tissues and organs, such as the lung¹, heart² as well as broader
48 cross-tissue atlases^{3,4}. However, single-cell RNA sequencing technologies have the intrinsic limitation of

49 requiring cells to be dissociated from their native tissue context. This results in a loss of information about the
50 cellular microenvironment, which leads to an incomplete picture of the molecular factors of variation^{5,6}.
51 Recent advances in image-based spatial transcriptomics technologies have enabled the measurements of
52 single-cell RNA-seq *in situ*, facilitating the profiling of hundreds of genes in hundreds of thousands of cells
53 across various tissues and organs^{6,7}. In situ spatial omics has been successfully applied to many biological
54 systems, tissues and organs, revealing spatial components of cellular variations such as cell-cell
55 communication⁸ and spatial gradients as well as emergent properties of tissue niches⁹, for example, in the
56 mouse and human brain^{10,11}, lung¹², and liver¹³. Here we hypothesize that spatial omics data is becoming rich
57 enough to learn a spatially aware, 'foundational' representation of cellular variation at scale.

58 A foundation model is a deep learning model trained on broad data that can be adapted to a wide range of
59 downstream tasks¹⁴. In recent years, foundation models have emerged as a game-changing modeling
60 paradigm, revolutionizing fields such as natural language processing^{15,16} and computer vision¹⁷. Foundation
61 models increasingly account for multimodal data, by leveraging not only one data modality, e.g. text, but also
62 images, video, and audio¹⁸. By leveraging massive datasets, powerful machine learning architectures, and
63 large compute resources, foundation models are able to learn general representations of language, vision, or
64 domain-specific data such as DNA^{19–21} and protein sequences^{22–24}, outperforming more classical modeling
65 approaches in a series of downstream tasks. Most commonly based on transformer architectures²⁵, these
66 models are pretrained on vast, unlabeled data, thus ditching the classical, costly tasks of sample annotation.
67 Through self-supervision²⁶, they learn powerful representations of complex data by identifying patterns and
68 relationships within the data itself without the need for human-labeled examples. These learned
69 representations then serve as a strong base for downstream tasks, while fine-tuning on labeled data further
70 enhances performance on specific applications.

71 The field of single-cell biology has taken up deep-learning-based representation learning for some time,
72 leveraging autoencoders^{27–29} for analysis tasks such as data integration³⁰, atlas mapping³¹ and perturbation
73 prediction^{32–35}. Recently, the field has witnessed a surge in the development of foundation models designed
74 explicitly for single-cell genomics data^{20,21,36–44}. These models differ in tokenization and learning strategies,
75 yet most of them leverage the transformer architecture with self-attention²⁵. They rely on large data
76 collections, usually in the order of tens of millions of cells, for pretraining^{21,37,38}. The representations of genes
77 and cells learned by these models are derived from implicitly modeling the complex interplay between gene
78 expression patterns within a single cell via the flexible transformer architecture. Single-cell foundation models
79 are evaluated on a variety of downstream tasks, such as cell type classification^{21,38}, gene regulatory network
80 inference^{21,37}, or prediction of cellular responses to perturbations^{37,40,42}. The diversity and complexity of these
81 tasks are useful to thoroughly probe the model's performance and to evaluate the robustness of the learned
82 representation and the model's ability to generalize to complex predictive tasks. Current results are
83 promising but not entirely replicated in independent benchmarks^{45–50}. Notably, to date, none of these models
84 account for spatial relationships of cells during training, with the exception of CellPLM⁴⁰, which, however, is
85 trained on a limited dataset of 9 million dissociated and 2 million spatial transcriptomics cells⁴⁰ and not
86 fine-tuned on spatial tasks beyond gene imputation.

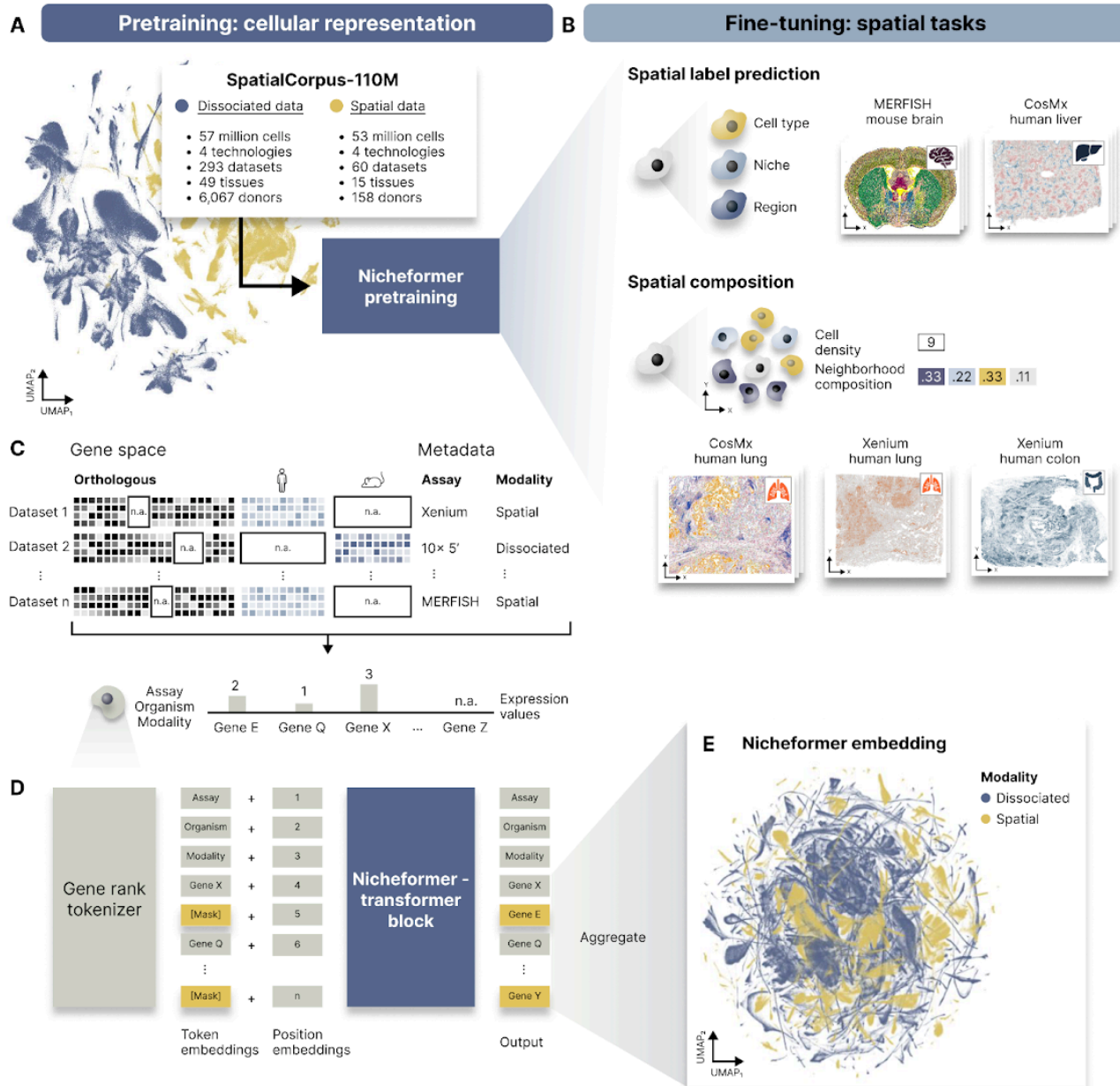
87 We propose Nicheformer, a novel spatial omics foundation model to understand tissue dependencies.
88 Nicheformer is pretrained on large-scale single-cell and spatial transcriptomics data to enable predictions for
89 spatially dependent tasks with limited data availability. To learn spatial cellular representation at scale, we
90 compiled the, to date, largest collection of single-cell and spatial transcriptomics datasets, spanning over 110
91 million cells from both human and mouse from 73 different organs and tissues. The Nicheformer data
92 collection is the first large-scale pretraining corpus that contains not only dissociated single-cell data but also
93 comprises over 53.83 million cells that were measured using image-based spatial technologies. By
94 incorporating contextual information through modality, organism, and assay tokens, Nicheformer is able to
95 learn a joint representation of single-cell and spatial genomics. We designed a set of novel downstream
96 tasks showing that both fine-tuned Nicheformer and a linear probing model trained on the Nicheformer
97 embedding systematically outperform existing embedding models like scVI^{27,28} and PCA for these tasks. We
98 demonstrate that Nicheformer accurately transfers the spatial context identified in spatial transcriptomics
99 onto dissociated single-cell data, allowing users to enrich classical single-cell RNA-seq data with spatial

context. This work paves the way for a new generation of foundation models for learning robust representations of cellular variation in tissues.

Results

A transformer-based foundation model for combined spatial and disassociated single-cell data

Overview. Nicheformer is a transformer-based model pretrained on a large curated transcriptomics corpus of dissociated and spatially resolved single-cell assays containing more than 110 million cells, which we refer to as SpatialCorpus-110M (Fig. 1A). Foundation model approaches build upon the idea of representing data in a way that allows a large language model, realized by the transformer architecture, to understand fine-grained dependencies in the data; the main differences lie in the way the data is encoded and how the model is trained and fine-tuned. Nicheformer's novelty lies in the encoding of sample covariates across technologies, species and modalities, which enables a general architecture and its use of downstream tasks via linear probing and uncertainties. We evaluated Nicheformer on a novel set of relevant downstream tasks that aim to demonstrate the model's capability to transfer spatially inferred cellular variation to single-cell dissociated data (Fig. 1B).



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Figure 1 | Nicheformer, a foundation model for spatial transcriptomics. A) Nicheformer is pretrained on the SpatialCorpus-110M, a large data collection of over 110 million cells measured with dissociated and image-based spatial transcriptomics technologies. The SpatialCorpus-110M collection comprises single-cell data from Homo Sapiens and Mus Musculus across 17 distinct organs, 18 cell lines, and additional single-cell data from other anatomical systems and junctions. Shown is an exemplary UMAP visualization of a random 1% subset of the entire pretraining dataset ($n=1,108,759$ cells) of the non-integrated $\log_2$ -transformed normalized SpatialCorpus-110M colored by modality. B) Nicheformer includes a novel set of downstream tasks, ranging from spatial cell type, niche and region label prediction to neighborhood cell density and neighborhood composition prediction. We test our approach on large-scale, high-quality spatial transcriptomics data from the brain (mouse - MERFISH), liver (CosMx - human), lung (CosMx - human, Xenium - human), and colon (Xenium - human). Visualized are example slices of the respective datasets colored by niche labels (brain, liver, and lung) and cell density (lung and colon). C) The SpatialCorpus-110M is harmonized and mapped to orthologous gene names, as well as human and mouse-specific genes, to create the input for Nicheformer pretraining. We harmonized metadata information across all datasets, capturing species, modality, and assay. D) Each cell's gene expression profile and metadata are fed into a gene rank tokenizer to obtain a tokenized representation for each cell. The tokenized cells serve as input for the Nicheformer transformer block to predict masked tokens. Finally, the Nicheformer embedding is generated by aggregating the gene tokens (Methods). E) The pretrained Nicheformer embedding is visualized as UMAP colored by modality. The UMAP shows a random 5% subsample of the entire Nicheformer embedding ($n=4,903,086$).

The Nicheformer pretraining corpus comprises transcriptomics data from both humans and mice (Fig. 1A). Only expression data was used during pretraining to train the model to integrate data from dissociated and targeted spatial technologies. Both technologies exhibit substantial technical batch effects (Fig. 1A). A limiting factor for image-based spatial transcriptomics data is the targeted feature space which only measures a few hundred to a few thousands of genes, depending on technology and panel^{51,52}. By pretraining across both modalities on the entire collection of datasets, instead of training organ-specific models, we aim to account for cross-tissue, cross-technology and cross-disease variations. To evaluate the representation learned in the Nicheformer pertaining, we focused for the downstream tasks on large-scale spatial transcriptomics datasets measured in four different solid organs with three image-based spatial transcriptomics technologies (Fig. 1B). We fine-tuned the pretrained Nicheformer model or performed linear probing on the pretrained Nicheformer embedding for each task and dataset separately. The linear probing resembles a zero-shot-like evaluation of the learned embedding. The embedding is obtained by calculating a forward pass for a specific dataset through the pretraining model to generate a lower-dimensional representation, the so-called Nicheformer embedding. This embedding is then used in a single-layer classifier or regression network for e.g. label prediction. The organ-specific spatial context learned by the fine-tuned or linear probing Nicheformer model can then be used to evaluate the model's ability to generalize information learned from spatial transcriptomics data, without directly accounting for the available spatial context, and transfer it to dissociated data.

Cell representation. We define a cell as a sequence of gene expression tokens whose order represents the level of expression of each gene. Since the SpatialCorpus-110M is composed of two different species, we represent their expression profiles by concatenating matched orthologous genes using just the protein-coding ones as well as genes uniquely present in either of the two species, resulting in an overall vocabulary size of 20,310 gene tokens (Fig. 1C, Methods). To sequentialize a single-cell expression vector, we need to rank individual gene measures in relation to the mean gene expression in the SpatialCorpus-110M. This way, each cell is converted into a sequence where each token is a gene, further referred to as a gene token, and those are ordered according to its relative expression (Fig. 1D, Methods). Encoding cells based on their gene ranks has been shown to generate embeddings that are robust to technical batch effects but preserve the gene-gene relationships in a cell²¹. We first concatenate all technology-specific datasets from the SpatialCorpus-110M and pad missing genes not measured for individual cells. Previous works^{52,53} have demonstrably shown technology-dependent biases between spatial and dissociated transcriptomics data. Since spatial transcriptomics measurements and the obtained signal heavily depend on the chosen preprocessing pipeline used to generate cell-by-gene count matrices⁵⁴, average expression profiles obtained with image-based spatial transcriptomics technologies can show much higher gene counts compared to dissociated technologies⁵². We argue that instead of calculating a non-zero mean vector across the entire SpatialCorpus-110M, it is preferred to compute a technology-specific non-zero mean vector, which accounts better for assay-specific gene expression effects. We therefore grouped all dissociated assays into one technology, whereas for the spatial transcriptomics datasets, we defined the overall technology categories as MERFISH, Xenium, CosMx and ISS. The technology-specific non-zero mean vector is then given by the mean expression value of each gene, without considering the zero counts. We additionally account for different modalities, technologies and species by adding contextual tokens that model the nuances of each of those elements. By incorporating the contextual tokens, the model is able to learn the nuances that differentiate organisms, modalities, and technologies in the measured cells.

Model design and training. The Nicheformer architecture uses a context length of 1,500 gene tokens serving as input for its transformer. The transformer block leverages 12 transformer encoder units^{16,25} with 16 attention heads per layer and a feed-forward network size of 1,024 to generate a 512-dimensional embedding of the pretraining dataset, resulting in altogether 49.3 parameters, which resulted in the best performance compared to models with fewer parameters (Supp. Fig. 1) and other model hyperparameters (Supp. Table 1). Nicheformer uses self-attention to weigh the importance of each gene in a cell in relation to every other gene, enabling it to capture gene-gene relationships effectively throughout pretraining. We confirmed technology-dependent biases between spatial and dissociated transcriptomics data by evaluating initial pretraining experiments that were trained on just a single modality (Supp. Fig. 2). We found that only training on dissociated single-cell data resulted in a lower performance in terms of F1 score for the task of

186 annotating cells in a spatial transcriptomics dataset in the brain. This indicates that dissociated data alone
187 can not resolve the sources of variation observed in spatial transcriptomics data.

188 **Model evaluation and downstream tasks.** Current transformer-based models designed for single-cell
189 genomics data are used for either gene-level tasks, such as inference of gene regulatory networks and
190 prediction of perturbation effects, or cell-level tasks, such as cell type annotation and batch integration^{21,37,38}.
191 By incorporating single-cell data at both the dissociated and the spatial scale into a single model, we are able
192 to define a novel set of biologically meaningful spatial tasks, where previous models primarily only focused
193 on disassociated ones (Supp. Table 2). These spatial tasks formulate biological questions related to the
194 cellular composition of tissues and are sufficiently challenging to resolve: they address the prediction of
195 human-annotated niche and region labels and spatial composition prediction (Fig. 1B, Methods). For the
196 spatial label prediction tasks, we also evaluate the model's uncertainty regarding the predicted labels
197 (Methods). For spatial composition tasks, we define a distance-based spatially homogeneous niche around
198 each cell and ask the model to predict local density or cell type composition. The tasks are formulated as
199 prediction problems operating on Nicheformers pretrained embedding (Fig. 1E). Importantly, the obtained
200 embedding is not comparable to commonly known integrated spaces as retrieved in single-cell atlases, but
201 rather captures a cross-technology, cross-tissue and cross-species cell representation that reflects
202 meaningful information usable for downstream tasks.

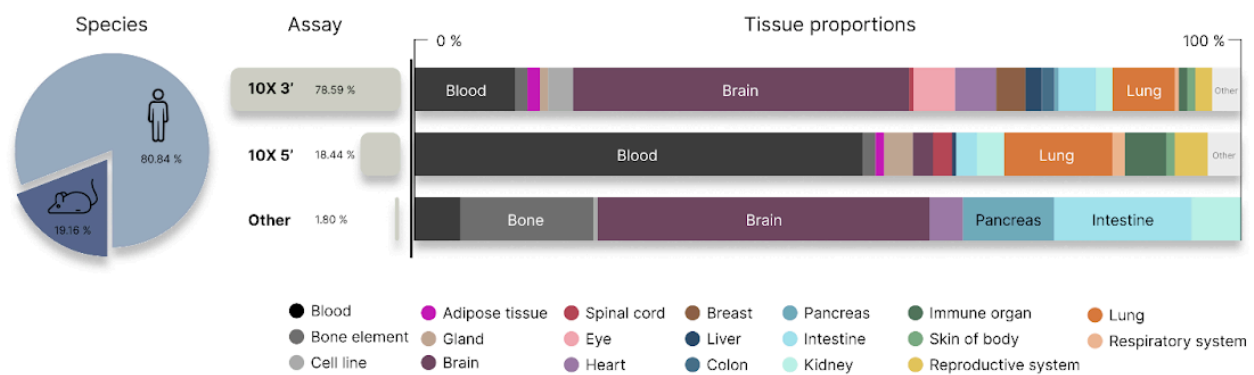
203 **Model transfer-learning.** We investigate the performance of Nicheformer in a zero-shot-like scenario using
204 linear probing as well as a fine-tuning setting. Both fine-tuning and linear probing involve adding a new linear
205 layer on top of the pretrained Nicheformer model adapted to the prediction task formulated above.
206 Fine-tuning also updates the parameters within the Nicheformer model itself, while linear probing keeps
207 those parameters frozen. Through the low complexity of the linear probing mode, we ensure that the
208 biological insights result from the meaningful biological information captured by the learned Nicheformer
209 embedding (Fig. 1E).

210 **SpatialCorpus-110M**, a large-scale, cross-organ and cross-species pretraining dataset for
211 single-cell and spatially resolved transcriptomics

212 To train the Nicheformer model, we collected and harmonized a new, large-scale pretraining corpus of
213 single-cell and spatially-resolved transcriptomics data from human and mouse tissues. To our knowledge,
214 this resource consists of the, to date, largest collection of harmonized and combined single-cell and spatially
215 resolved transcriptomics datasets.

216 SpatialCorpus-110M consists of a total of 57.06 million cells from single-cell dissociated technologies. It
217 builds upon the CellXGene CENSUS database (33.47 million cells), which we extended by an additional 180
218 datasets across 73 different tissues, containing 17 solid organs, 18 cell lines and various additional tissue
219 junctions in human and mice, with harmonized ontologies and metadata (Fig. 2A). These additional
220 dissociated datasets have been collected through Gene Expression Omnibus^{55,56}, sfaira⁵⁷ and the Human
221 Cell Atlas data explorer⁵⁸ (Supp. Table 3, Methods).

A Dissociated collection - 57.06 million cells



B Spatial collection - 53.83 million cells

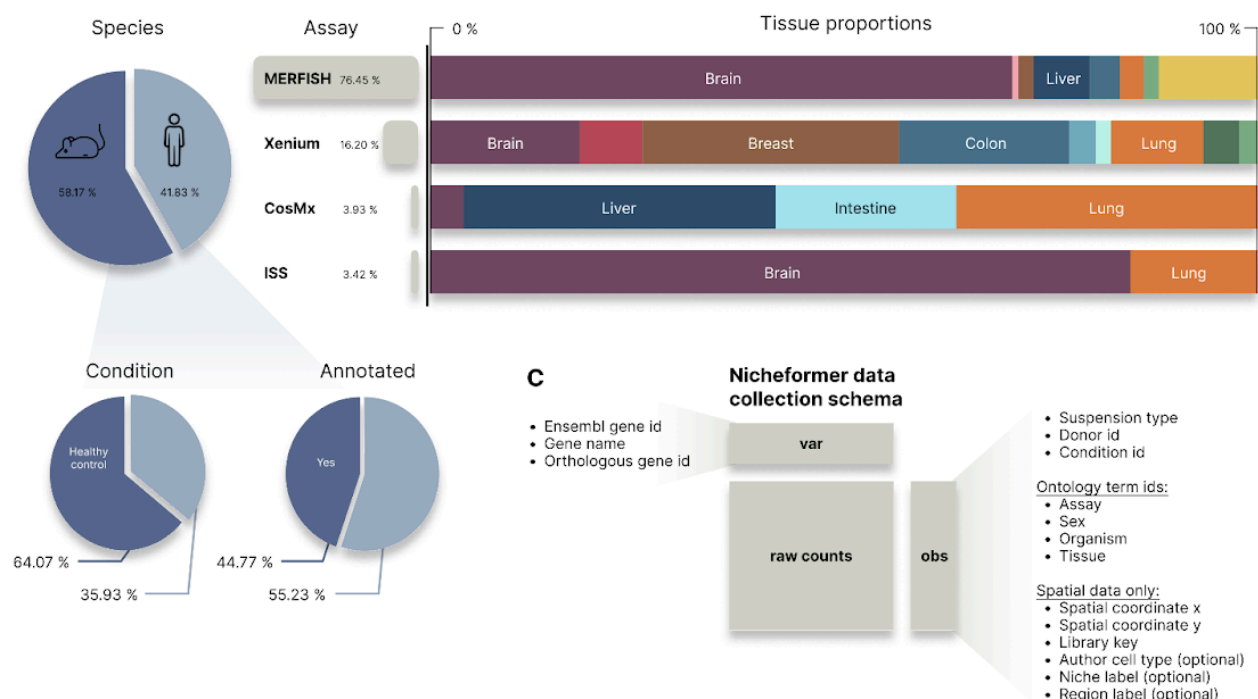


Figure 2 | Overview of the SpatialCorpus-110M collected for training Nicheformer. A) The dissociated single-cell genomics data collection contains 57.06 million human and mouse cells. The collection includes cells from 17 different organs, 18 different cell lines, blood, bone elements, tissue junctions, and other anatomical entities, grouped by primary solid organ to simplify visualization and analysis. B) The spatial transcriptomics data collection contains 53.83 million targeted spatially-resolved cells obtained from humans and mice. The collection comprises four different profiling technologies across 15 different solid organs. C) The SpatialCollection-110M was collected with harmonized metadata defined in the Nicheformer data collection schema (Methods). Metadata was harmonized both on gene and on cell level depending on modality.

Altogether, the dissociated collection of SpatialCorpus-110M comprises cells from over six thousand different donors and technical or biological replicates. For spatial transcriptomics, we curated image-based spatial datasets, specifically MERFISH⁵⁹ (Vizgen MERSCOPE), 10x Genomics Xenium, Nanostring CosMx¹², and In Situ Sequencing (ISS)⁶⁰ data (Fig. 2B, Supp. Table 4). Data was collected from individual publications as well as via the Vizgen data release⁶¹ (18.8%) and the 10x Genomics data resource⁶² (13.7%). Altogether, the spatial data consists of 53.8 million cells across 15 tissues in human and mouse, including 158 individuals or animals and over 10,600 individual tissue sections. The biggest proportion of cells in the spatial collection originated from the brain, with 60.46% (n=32,146,779 cells) of all cells collected, followed by the lung with 9.95% (n=3,199,548 cells). A large proportion of the publicly available spatial omics datasets we collected is not annotated (55.23%). Cells are mostly from healthy controls (64.07%), but we also included cells from

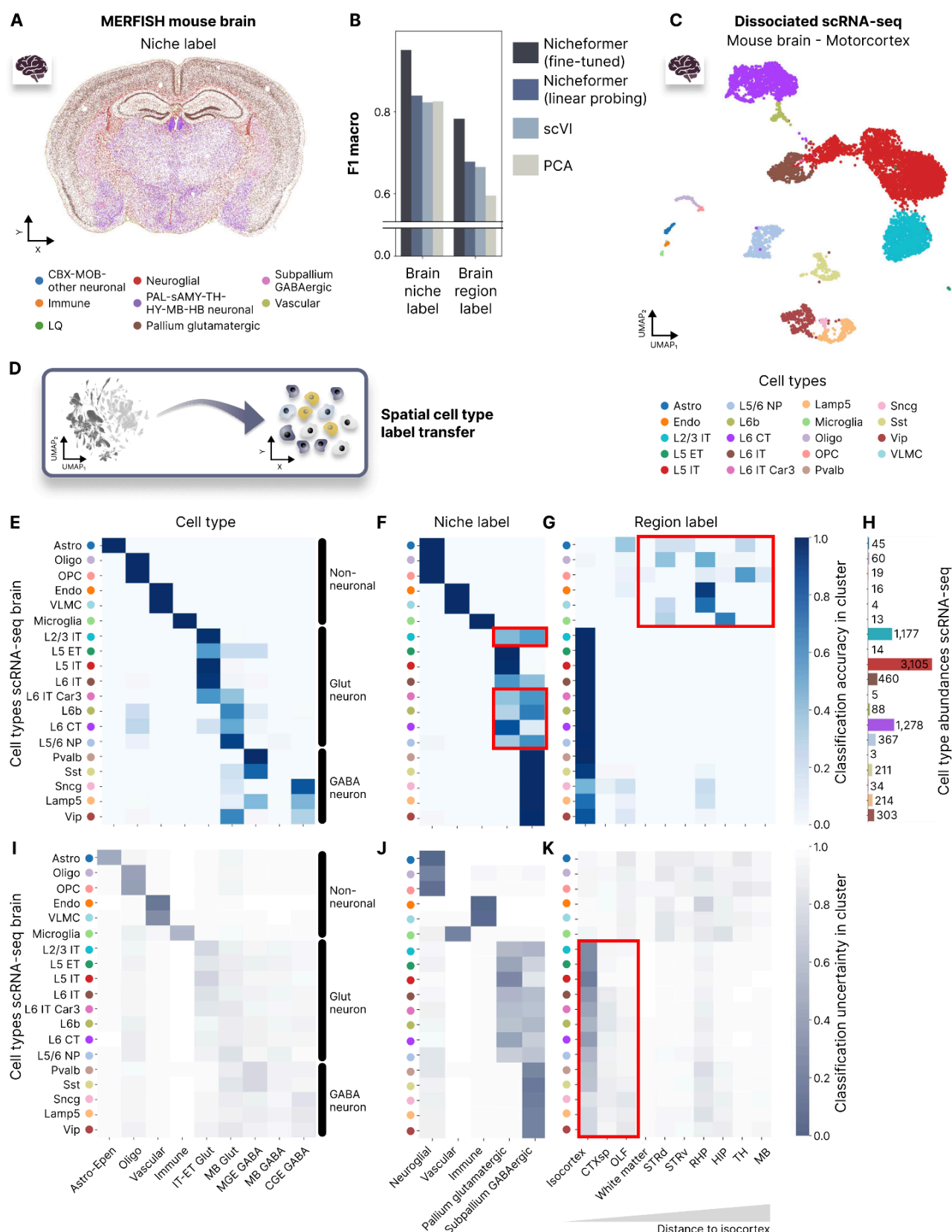
various cancer samples (31.98%) to enable Nicheformer to learn meaningful representations of the tumor-immune microenvironment in space.

For all datasets in the Spatial-Corpus-110M, we curated metadata, such as assay, sex, organism, and tissue, based on the original publications by using official ontology term identifiers (Fig. 2C, Methods). To harmonize features across species, tissues and assays, we first converted all gene symbols to ENSEMBL gene IDs using pyEnsemble⁶³. Then we used BioMart⁶⁴ through the official Ensembl releases⁶⁵ to match orthologous genes between species. This resulted in 16,981 orthologous genes matched between mouse and human, 151 genes specific only to mouse and 3,178 genes specific only to human. This harmonization generated a vocabulary size of 20,310 unique gene ID tokens.

Nicheformer allows transferring spatially-resolved cell type, niche and region labels onto unseen data

Dissociated single-cell atlases excel at capturing molecular maps of the cell type diversity of tissues and organs¹. Cells are typically well described by their cell type, commonly defined as a group of cells that share a characteristic and stable molecular state, reflected in their structure and function⁶⁶. These cell types can be found in one specific tissue or across multiple tissues⁴. Typically, cell types are defined without directly incorporating the cellular microenvironment⁶⁷. However, a cell's spatial context provides additional value and information beyond the cell's expression profile and can be helpful for understanding cellular microenvironments⁶⁸. Spatially-resolved single-cell genomics allows us to augment the concept of cell types when considering also the molecular state of the neighbors of a cell. We refer to this as a cell niche. In contrast to standard approaches to identifying cell types, cellular niches are defined by spatially-dependent labels obtained by combining similarity in gene expression profiles together with spatial neighborhood structures, histological information or global structures in the tissue^{9,69–71}. Cellular niches typically describe a local structure in a tissue, for example, a tumor niche or immune niche, which vary both in terms of gene expression, but also in terms of the cell type compositions. Similarly to niches, tissue regions can be defined as spatially consistent cellular states across a larger-than-a-niche tissue context. Importantly tissue regions are commonly composed of multiple different niches and therefore describe higher-level structures in a tissue.

To test whether Nicheformer, pretrained only on gene expression information, can efficiently transfer spatial context with minimal to no supervision we considered a large MERFISH mouse brain dataset¹⁰, where 17 different brain regions were defined by registering a 10×10 μm grid of cells assigned to their anatomical parcellation of the Allen Mouse Brain Common Coordinate Framework⁷² v3 (Suppl. Fig. 3A). The resulting CCFv3 registration was then used to obtain a region label for each cell measured in the MERFISH mouse brain dataset¹⁰. This dataset additionally consists of seven distinct tissue niches (Fig. 3A) defined based on their molecular and anatomical similarity, which are structurally organized across the entire brain (Suppl. Fig. 3B).



Nicheformer accurately classifies cells from the dissociated motor cortex to relevant cell types (n=9 out of 33 distinct ones in the classifier) trained on the whole mouse brain MERFISH dataset. F-G) Nicheformer correctly projects dissociated single-cells to niche (F) and region (G) labels to provide spatially dependent labels. F) Nicheformer misclassified parts of L2/3 IT neurons as residing in the subpallium GABAergic niche (highlighted in the red box). Additionally, the deep cortical excitatory neurons L6b, L6 CT, L6 IT, and L6 IT Car3 (highlighted in the red box) should be classified as pallium glutamatergic niche instead of subpallium GABAergic by Nicheformer. G) Most of the non-neuronal cells (84.7 % of all non-neuronal cells, n=133) were misclassified as not belonging to the isocortex or the adjacent brain regions (highlighted in the red box). H) Cell type abundances in the scRNA-seq dataset measuring the primary motor cortex in the mouse. I-K) Classification uncertainty of label transfer of the dissociated scRNA-seq dataset to the MERFISH mouse brain data for cell type label (I), niche label (J), and region label (K) with a value of 0 representing a high uncertainty and 1 being a lower uncertainty, i.e., high certainty. K) Observed high uncertainty for parts of the Glut and GABA neurons for the region prediction of the isocortex, CTXsp and OLF, which are neighboring brain regions.

In this setting, we assessed Nicheformer's ability to accurately impute niche and region labels on unseen, held-out tissue sections from the MERFISH mouse brain dataset, measuring one male mouse brain (Suppl. Fig. 3A-D). We evaluated two versions of Nicheformer, a zero-shot-like use of the pretrained embeddings (Suppl. Fig. 4A) via linear probing as well as fine-tuning the model for the prediction task. To assess model performance, we ask whether the Nicheformer embedding better captures information required for the downstream niche and region prediction task compared to more classical embeddings calculated with a variational autoencoder (scVI) and PCA on the specific data set without pretraining - in dissociated tasks it has proven to difficult for foundation models to outperform data set specific embeddings. First, we trained scVI and PCA on the training set of the MERFISH mouse brain dataset and then generated an embedding for all cells present in the dataset (Methods). We then used the resulting embeddings as input for a linear classifier in a similar way as done for the Nicheformer representation (linear probing).

We found that both of our approaches outperform traditional embedding methods, scVI and PCA, in terms of macro F1 score, which effectively balances precision and recall across all classes (Fig. 3B). Notably, while the linear probing approach consistently outperforms the baselines, fine-tuning Nicheformer further enhances performance, demonstrating the capability of the model pretrained only on gene expression information to adapt to spatial tasks. We performed a similar analysis on a randomly held-out test set of the CosMx human liver dataset defining tissue niches as different zonations between the central and portal veins (Suppl. Fig. 5A-C). Again, the fine-tuned Nicheformer model showed the best performance in terms of macro F1 score. However, the Nicheformer linear probing model did not outperform linear probing models trained on scVI and PCA (Suppl. Fig. 5F). We hypothesized that the worse linear probing performance is related to the insufficient model capacity due to limitations in the number of trainable parameters and training time. This prevents the model from accurately learning a nuanced representation of liver cells, which have relatively low overall abundance in the SpatialCorpus-110M (Fig. 2A, B). We, therefore, asked whether additional pretraining of Nicheformer solely on the liver training set improved performance. Indeed, we observed an improvement in linear probing performance for the prediction of the liver niche labels on the longer pretrained Nicheformer embedding (Suppl. Fig. 5F). We conclude that longer pretraining can be beneficial for datasets with lower abundance in the training corpus.

Next, we assessed whether Nicheformer accurately transfers cell type labels defined in the MERFISH mouse brain dataset onto dissociated cells present in the SpatialCorpus-110M and measured in the primary motor cortex cells with scRNA-seq (Fig. 3 C, D)¹¹. We find that Nicheformer correctly selects the nine motor-cortex related cell types out of the overall 33 cell types present in the MERFISH mouse brain dataset (Fig. 3E, Suppl. Fig. 6A). When calculating classification uncertainty based on the overall predicted distribution generated by the model (Methods), the predicted cell type labels show overall a high agreement and low classification uncertainty (Fig. 3E, I) with the original cell type annotations. Nicheformer correctly assigns all non-neuronal cells to their mapped spatial cell type (Fig. 3E, Methods). Glutamatergic (Glut) neurons, which are the most abundant cell types in the dissociated dataset (Fig. 3H), are also correctly mapped to the correct neuronal identity (Glut); however, the regionalization for a few Glut neuronal subtypes from deeper cortical layers (L6b, L6 CT, L5/6 NP) is mispredicted as midbrain glutamatergic (MB Glut), while one would expect an assignment to NP-CT-L6b Glut (Suppl. Fig. 6A). The misclassification of the region of the glutamatergic neurons can be linked to the overall transcriptional diversity in the fine-tuning MERFISH mouse brain dataset. MB Glut cells are sub-clustered into 657 subtypes, whereas NP-CT-L6b Glut cells only

define 83 distinct subtypes even though they are overall more abundant in the MERFISH mouse brain dataset (MB Glut - n=88,169 cells, NP-CT-L6b Glut - n=174,616 cells)¹⁰.

Next, we wanted to assess whether Nicheformer is able to annotate brain niche labels in the same scRNA-seq primary motor cortex dataset. We find that Nicheformer correctly predicts only the expected niche labels for all non-neuronal cell types with low uncertainty (Fig. 3F, J, Suppl. Fig. 6B). Additionally, all GABAergic (GABA) and Glutamatergic (Glut) neurons are correctly predicted to be part of either the pallium glutamatergic or subpallium GABAergic (Fig. 3F), yet with higher uncertainty for excitatory neuronal subtypes (Fig. 3J). However, we also observe a partial misclassification for parts of the Glut neurons, which is likely due to the spatial overlap of the pallium glutamatergic and subpallium GABAergic niches (Suppl. Fig. 6D, E) and the high diversity of cell types within the pallium glutamatergic and GABAergic niches.

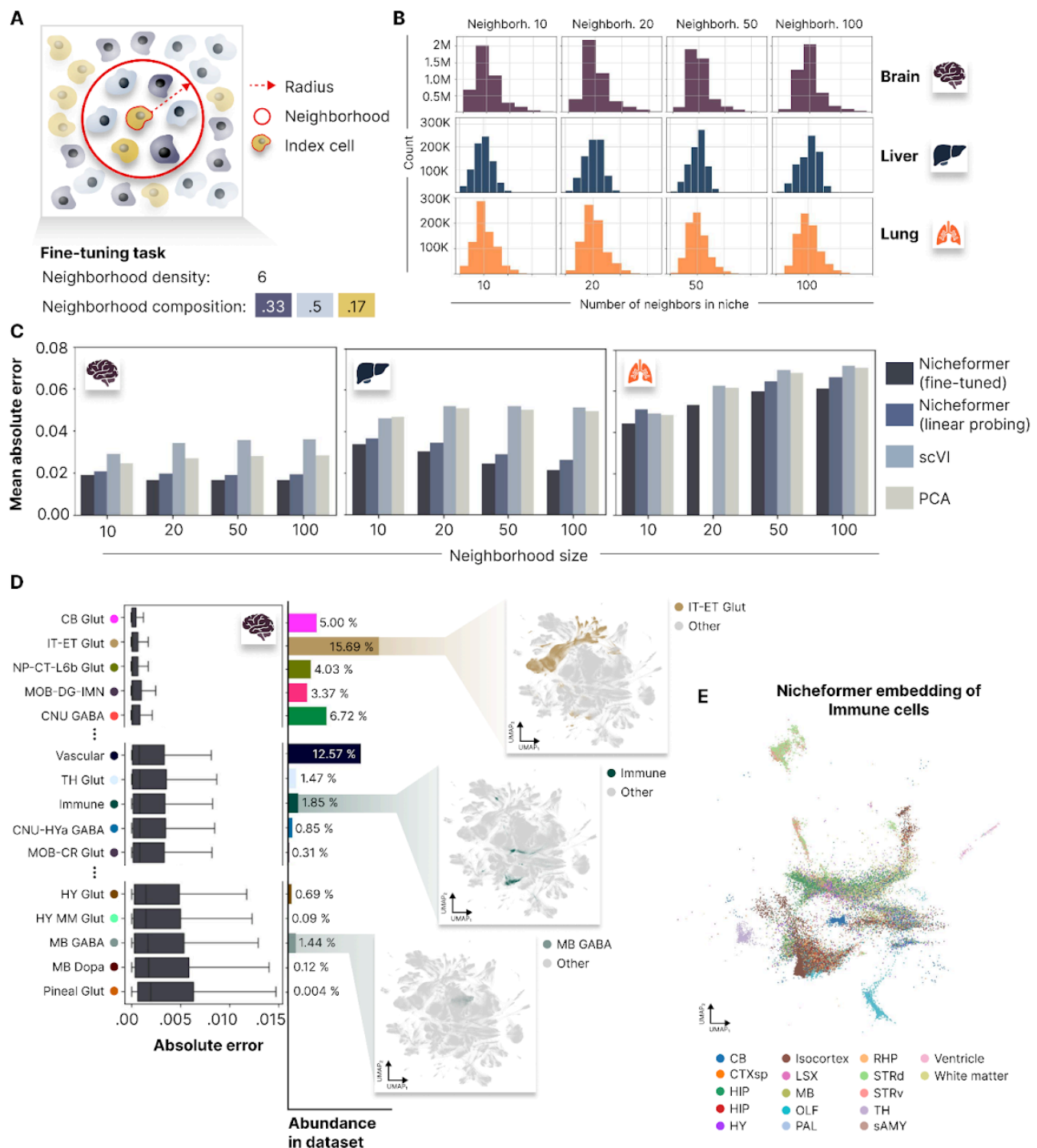
We then assessed whether Nicheformer correctly annotates brain regions to the primary motor cortex dissociated cells. The expected region for all cells is the isocortex. Nicheformer correctly predicts the isocortex for all neurons, with very low uncertainty for the excitatory neurons and slightly higher uncertainty for the inhibitory neurons (Fig. 3G, K, Suppl. Fig. 6C). We also observe that some cells are classified in brain regions neighboring the isocortex, namely the cortical subplate (CTXsp), olfactory area (OLF), and white matter (Fig. 3G, K). This classification might be plausible as during tissue dissection other brain regions might be additionally captured. For region classification, we observed a lower performance compared to cell type and niche label for the non-neuronal cells. This could be related to the lower transcriptional diversity and their lower regional specificity compared to neuronal cells¹⁰.

Altogether, this demonstrates Nicheformer's ability to learn powerful cell representations by capturing nuanced spatial information. Linear probing on the pretrained model already surpasses existing baselines, highlighting the effectiveness of the representation. Fine-tuning further refines this representation, emphasizing the importance of task-specific adaptation for capturing subtle cellular variations. Notably, Nicheformer enables the direct transfer of spatially-aware annotations from spatial to dissociated single-cell data by using a simple linear layer. This capability unlocks new possibilities for analyzing single-cell data across different modalities.

Nicheformer predicts neighborhood compositions in spatial and dissociated single-cell data

Tissue microenvironments consist of cellular neighborhoods with a diverse composition of cell types. Differences in neighborhood composition have been shown to have a significant effect on gene expression and can be associated with cell-cell communication events⁸. Furthermore, the cellular composition of neighborhoods in the tumor microenvironment may hold prognostic value^{73,74}, since immune cell infiltration in the spatial context is a predictor for cancer survival^{75,76}. Here we show that we can leverage Nicheformer's multimodal cell representation to accurately relate changes in gene expression to differences in neighborhood compositions in spatial data and transfer them to dissociated transcriptomes.

We define a cell's 'computational' neighborhood as the set of cells within a fixed radius (Fig. 4A, Methods). The total number of cells composing the neighborhood defines the neighborhood density, and the proportion of cell types in the neighborhood defines the neighborhood composition. This notion is consistent with previous approaches defining a cellular neighborhood⁷⁷ and allows for an interpretable evaluation of model results. Generally, the definition of a cell neighborhood can be extended in the future to account for non-isotropic cell neighborhoods that might vary in their cell type composition and are drivers of similar biological functions with varying sizes across a dataset.



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Figure 4 | Nicheformer accurately predicts neighborhood compositions at multiple niche resolutions for the brain, liver, and lung. A) We define the neighborhood of a cell as its local neighborhood given a radius and an index cell. The neighborhood cell density is then defined by the number of cells in the neighborhood, and the neighborhood compositions are the proportions of neighboring cell types. B) Neighborhoods are computed at multiple resolutions resulting in different neighborhood size distributions. Each barplot shows the distribution of the number of neighbors across the brain, liver, and lung datasets. We extract neighborhoods with the mean number of neighbors 10, 20, 50, and 100 for each dataset. C) The fine-tuned and linear probing Nicheformer models outperform zero-shot models trained on scVI and PCA embeddings in terms of mean absolute error across all neighborhood sizes and all three organs, the brain, liver, and lung. D) Left: Fine-tuned Nicheformer performance on the MERFISH mouse brain data grouped by index cell type. Shown are the absolute error values between predicted and observed neighborhood composition vectors for held-out test cells. For each box in (D), the centerline defines the median, the height of the box is given by the interquartile range (IQR), the whiskers are given by $1.5 \times \text{IQR}$, and outliers are given as points beyond the minimum or maximum whisker. Center: Index cell type abundances in the entire MERFISH mouse brain dataset. Right: UMAPs of

391 MERFISH mouse brain Nicheformer embedding with the selected index cell type as color superimposed. E) UMAP of the
392 Nicheformer embedding of all immune cells in the MERFISH mouse brain dataset with region label as color
393 superimposed.

394 To evaluate Nicheformer's ability to predict neighborhood composition, we focused on three datasets
395 measuring three organs with two different technologies, namely MERFISH mouse brain, CosMx human liver,
396 and CosMx human lung. We computed neighborhood compositions at varying resolutions for each of the
397 three datasets separately. The radii were selected to contain, on average 10, 20, 50, or 100 neighbors
398 (Fig. 4B, Methods). We evaluated Nicheformer both in linear probing and fine-tuned settings for each dataset
399 and each neighborhood size individually and compared its performance to linear probing on embeddings
400 computed with scVI and PCA. We found that fine-tuned Nicheformer systematically outperforms the linear
401 probing models trained on Nicheformer embedding, scVI and PCA for this task on all three organs in terms of
402 mean absolute error (Fig. 4C). Interestingly, Nicheformer's performance increased with neighborhood size in
403 the case of the brain and liver dataset. In the liver, we observed a stronger performance trend which might be
404 related to transcriptional patterns of zonation and structural components in the liver⁷⁸. For the CosMx liver
405 dataset, we additionally evaluated whether a multi-task multi-layer perceptron (MLP) would allow the
406 prediction of all neighborhood sizes jointly (Methods). We observed that a multi-task MLP did not outperform
407 a neighborhood size-specific linear probing model or the fine-tuned Nicheformer model, indicating that
408 downstream tasks should be evaluated separately (Suppl. Figure 5G).

409 To understand the model's behavior and performance in more detail, we additionally assessed the fine-tuned
410 Nicheformer performance for each cell type separately in the MERFISH mouse brain dataset (Fig. 4D,
411 Methods). We computed the absolute error between predicted and true neighborhood compositions across
412 all four neighborhood sizes and sorted the result based on the median values per cell type. We found that
413 the most accurately predicted cell types in terms of absolute error are also within the eight (out of 33) most
414 abundant cell types in the MERFISH mouse brain dataset. In contrast, the four cell types for which
415 Nicheformer performed worse are in the 14 least abundant cell types (Fig. 4D). For example, highly
416 abundant cell types predominantly from cortical layers (IT-ET Glut, NP-CT-L6b Glut) are structurally
417 organized in the brain and have a quite homogeneous neighborhood composition. Those two factors help to
418 explain the very accurate Nicheformer predictions. Similarly, CB Glut cells are based in the cerebellum, an
419 area with very high cell density⁷⁹ and high neighborhood homogeneity. Even though they have a lower
420 abundance in the overall dataset, Nicheformer accurately predicted their neighborhood composition
421 (Fig. 4D). On the other hand, Nicheformer shows a lower performance on cell types predominantly found in
422 the midbrain or hypothalamus (MB GABA, MB, Dopa, HY Glut, Hy MM Glut). These cell types are relatively
423 rare cell types in the given dataset and are located in more diverse and complex tissue layouts and show a
424 greater variety of neighboring cell types^{10,79}. This indicates that regionally diverse and less abundant cell
425 types in the pretraining corpus are harder to predict for the Nicheformer model. The performance differences
426 might be related to the structural properties of the brain regions as well as their varying cell type
427 compositions and abundance in the dataset. We further observed a relatively good performance of
428 Nicheformer for the neighborhood composition prediction of immune cells, despite their relatively low
429 abundance and their lack of regional specificity in the brain. Immune cells are scattered across the brain and
430 accomplish very specific but differing tasks ranging from regulating synaptic plasticity, and immune
431 surveillance, to preventing excitotoxicity⁸⁰. Interestingly, the Nicheformer embedding of the immune cells in
432 the MERFISH mouse brain data preserves the region information of those cells and region-specific
433 subclusters can be identified (Fig. 4E).

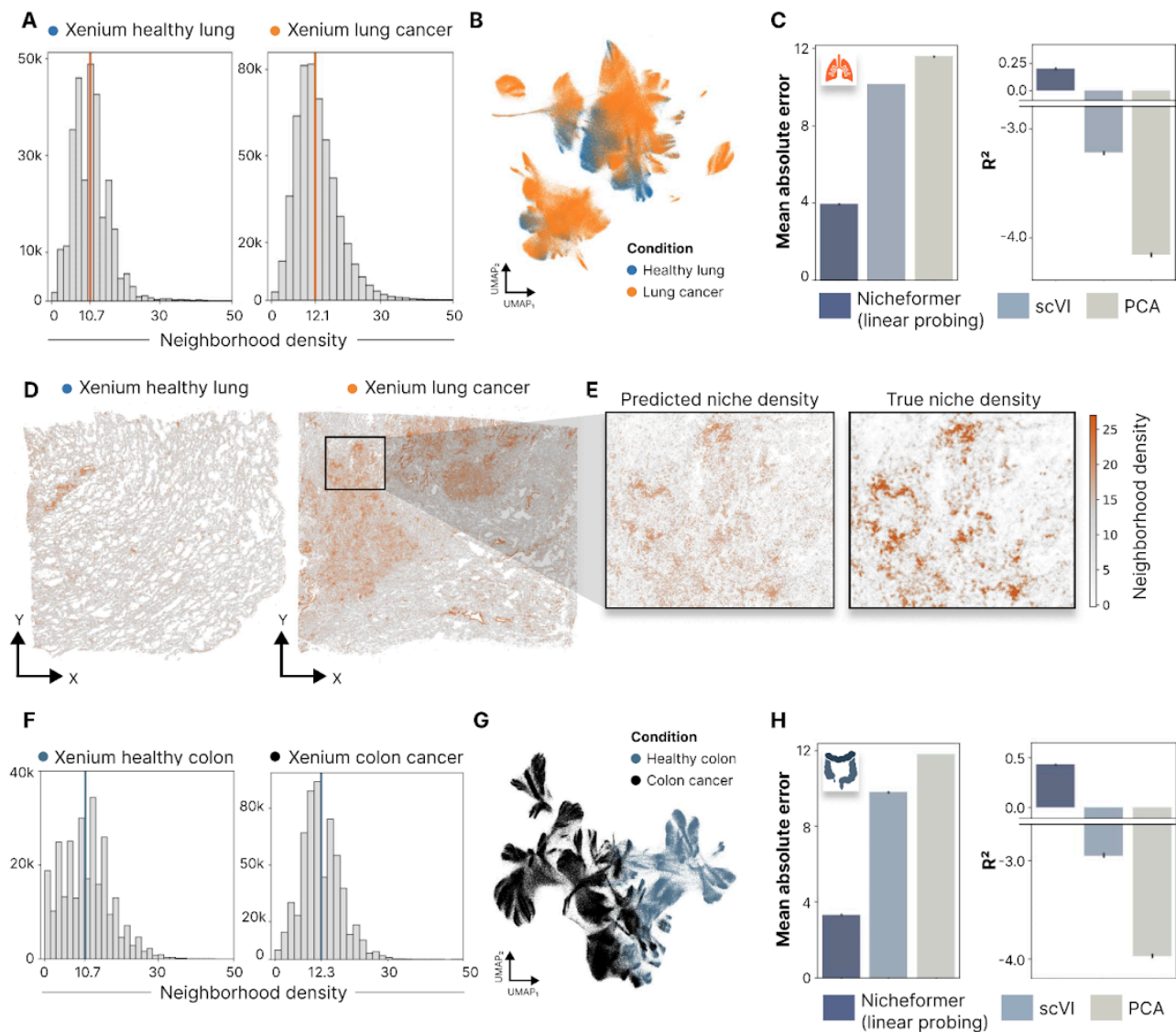
434 To assess whether our results generalize across organs and technologies, we performed a similar analysis
435 for the CosMx human liver dataset, evaluating the overall cell type performance in the task of predicting the
436 neighborhood composition across resolutions (Suppl. Fig. 5H). Again, we observed that Nicheformer's
437 performance heavily depends on the cell type abundance in the dataset and the regional specificity of the
438 individual cells, e.g., we saw a lower absolute error for hepatocytes compared to circulating immune cells
439 (Suppl. Fig. 5H). Hepatocytes are predominantly found in highly structured cellular microenvironments and
440 show strong spatial patterns in their gene expression⁸¹. While liver-resident immune cell populations were
441 shown to be mobilized under certain circumstances, hence their regional specificity might be lower compared
442 to other cell types⁸². This indicates that the Nicheformer embeddings can be useful to identify and

understand region- and niche-specific structures and differentiate cell types that show a higher regional specificity.

Nicheformer infers cellular niche density in unseen data

Beyond cellular niche labels and neighborhood composition, we asked whether local cell density is encoded in a cell's expression profile. It is long known that cell density can strongly affect growth behavior in vivo and in culture; also, increased cell density is a key feature of the formation of the tumor microenvironment, which leads to the creation of a hypoxic environment and depletion of infiltrating immune cell populations^{83,84}. For example, in colon cancer, it was shown that the immune cell density is associated with patient survival and can be used for tumor-immune patient stratification for improved anticancer therapy⁷⁵. In non-small cell lung cancer⁷⁴, immune cell density and neighborhood compositions were used to stratify specimens into groups associated with clinical outcomes.

We tested whether Nicheformer accurately predicts the neighborhood density in a Xenium lung dataset measuring an adult human healthy lung section and a section with invasive adenocarcinoma from a second patient⁸⁵, and in a Xenium FFPE-preserved healthy and diseased colon with stage 2A adenocarcinoma from two different patients⁸⁵. Consistent with literature observations^{75,74}, we observed a higher average cellular density in the cancer sections (colon=12.3 cells, lung=12.1 cells) compared to healthy tissue (colon=10.7 cells, lung=10.7 cells) when extracting cellular neighborhoods at the same radius (Fig. 5A, F, Methods).



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Figure 5 | Nicheformer accurately predicts changes in cellular neighborhood density in the lung and colon. A) Barplot of cellular neighborhood densities split by condition for the Xenium human lung dataset. B) UMAP of the Nicheformer embedding of the Xenium human lung dataset colored by condition. C) Mean absolute error and R^2 for the cellular neighborhood density prediction task for a Nicheformer linear probing model and linear probing models trained on scVI and PCA embeddings. D) Spatial allocation of cells in the Xenium human lung dataset colored by predicted cellular neighborhood density in the healthy and diseased lung. E) Predicted versus true cellular neighborhood density for a zoomed-in section of the Xenium lung cancer section. F) Barplot of cellular neighborhood densities split by condition for the Xenium human colon dataset. G) UMAP of the Nicheformer embedding of the Xenium human colon dataset colored by condition. H) Mean absolute error and R^2 for the cellular neighborhood density prediction task for a Nicheformer linear probing model and linear probing models trained on scVI and PCA embeddings.

We first computed Nicheformer embeddings for both datasets by generating a forward pass through the Nicheformer pretrained model (Fig. 5B, G). Additionally, we embedded the two datasets with scVI, and PCA (Methods). The three resulting embeddings for the datasets are then used as input for a linear probing regression model to predict the cellular neighborhood density for each cell. The linear probing models trained on the scVI and PCA embeddings failed to correctly predict the mean density and performed worse than random prediction, resulting in negative R^2 -values for both tissues. Interestingly, the linear probing model trained on the Nicheformer embedding outperformed the other two models in terms of mean absolute error and R^2 (Fig. 5C, H) and was able to accurately predict a higher cellular density in the tumor regions and denser tissue structures in the Xenium lung dataset (Fig. 5D). This demonstrates that the Nicheformer embeddings are able to capture neighborhood density variation solely on transcriptome information better than the baselines. Nicheformer's ability to infer cellular neighborhood density in healthy and cancer tissue can be useful to inject spatial relationship information in dissociated data to further characterize cell state variation in systems such as the tumor microenvironment.

Discussion

Nicheformer demonstrates the potential of multiscale foundation models for dissociated single-cell and spatial transcriptomics data. By leveraging the SpatialCorpus-110M and evaluating the model in different spatially informed downstream tasks and assessing the model's prediction uncertainty, we demonstrate that Nicheformer captures complex relationships between gene expression and spatial context. We introduce a novel set of downstream tasks designed explicitly for spatial data analysis, in which Nicheformer consistently outperforms baseline models, highlighting its effectiveness in learning a cell representation that is able to predict spatial features. These results strongly suggest that spatial context can be effectively inferred from transcriptomics data using Nicheformer. Further, Nicheformer paves the way for transferring spatial information to large collections of dissociated single-cell data which opens the door for more nuanced analyses of cellular function in tissue environment in silico.

A cell integrates its spatial context, i.e., its cellular neighborhood by cell communication, which is reflected in the cell's transcriptomic profile⁸⁶. This property has been used successfully to learn cell type communication profiles from coexpressed receptor-ligand interactions^{87–89}, to reconstruct spatial gene expression from spatial context and anchor points using optimal transport^{90–92} and to determine cell interactions beyond known receptor-ligands via graph neural networks^{77,93}. With Nicheformer, we build upon these results and show that we can predict spatial context from a cell's gene expression profiles alone with consistent accuracy. We found that, for example, immune cell neighborhoods in the brain are most likely encoded in the gene expression profiles, making it easier for Nicheformer to understand these differences and relate them to neighborhood composition changes. Extending this analysis to additional tissues will enable the characterization of recurrent immune niches across tissues and organs.

A long-term vision in systems biology has been to create multiscale models, from molecules and cells up to tissue, organs and eventually the whole organism^{94,95}. Nicheformer is a step towards creating a generalizable multiscale model for single-cell and spatial biology, bridging the gap from the single-cell to the tissue modality. More generally, it will be necessary to operate on multimodal data to generate a true representation of the cellular state. While spatial transcriptomics captures the cellular microenvironment in tissues well, integrating additional data modalities, such as protein abundance or epigenetic modifications, will provide a more complete picture of the cellular state. The development of multimodal foundation models faces multiple

challenges. One key hurdle is the lack of sufficient paired data measured across multiple or even all cellular modalities. However, with the development of new assays and sequencing technologies, we expect the number of multimodal datasets to grow, enabling the development of architectures to model them. Incorporating additional modalities will remain a challenge in the future as e.g. epigenetic modifications, protein abundance, or gene expression all have unique characteristics, and effectively combining them in a way that leverages their strengths remains an ongoing research area.

While Nicheformer represents a milestone for learning general representations for single-cell biology, we acknowledge some of the limitations of this approach. Firstly, Nicheformer performance depends on the data abundance and transcriptional diversity of the cells under study. Indeed, we showed that Nicheformer's performance for accurately predicting spatial labels and spatial compositions is impacted by cell and tissue type abundance in a spatial transcriptomics dataset. With the ongoing growth in spatial transcriptomics data availability as well as improved throughput thanks to technological advances, we expect that the prediction performance will increase across evaluated tissues. Secondly, Nicheformer does not explicitly incorporate the physical location of a cell during pretraining, limiting its capability to fully leverage the available information on spatial context. We deliberately chose not to include spatial coordinates during pretraining because we wanted to learn a general representation of gene expression variation across both modalities, fully supervised by gene expression alone. Nevertheless, we anticipate that future iterations of Nicheformer will account for spatial relationships of cells by encoding spatial neighbor graphs, for example, and potentially leveraging graph transformer architectures⁹⁶⁻⁹⁸ for the pretraining stage on spatial transcriptomics data. Graph transformers excel at modeling relationships between nodes in graphs, making them ideal for capturing nearest-neighbor effects on a cell's transcriptome. Thirdly, the interpretability of the Nicheformer model has not been fully explored. In future iterations, it would be interesting to inspect the learned architecture in order to understand interactions between genes within cells and niches to extract biological mechanistic knowledge, for example, by assessing how gene relationships are associated with cell state across the two modalities under consideration. We additionally see a need to scale Nicheformer in the number of parameters, pretraining time and dataset size. Characterizing scaling laws for foundation models in genomics has the potential to identify bottlenecks in learning schemes and datasets, thus informing design and pretraining choices for the next generation of models. Finally, we want to highlight the need for more comprehensive benchmarks than the set of spatial tasks presented here, which will help judge extensions and future alternative models. The field of biological foundation models is a novel area brimming with potential. However, unlike more established AI domains^{99,100}, there's a crucial gap in the form of standardized benchmarks for evaluating these models. Establishing robust benchmarks is a critical next step to compare and improve performance, rigorously assess methodological progress and guide future model development to identify the full potential of foundation models for single-cell biology.

Overall, Nicheformer demonstrates the feasibility of learning a foundational representation able to effectively transfer information from single-cell to spatial genomics and its reverse, paving the way for the next generation of foundation models trained on large heterogeneous collections of dissociated and single-cell data. We describe a set of new evaluations that are explicitly designed to probe the model's ability to encode spatial context and its transferability to a different modality that can be leveraged as a new benchmark for multimodal foundation models for single-cell and spatial genomics. We believe Nicheformer represents an important milestone towards building a general and robust representation of cellular biology phenotypes advancing our understanding of the heterogeneous effects of cellular niches in development and disease. We envision Nicheformer and similar models to actively assist in experimental design through hypothesis generation and experiment selection, ultimately accelerating the pace of scientific progress by helping to choose the next set of most informative experiments. Nicheformer will thus help to guide and design spatial experiments based on single-cell RNA-seq measurements, supporting the upcoming transition from cell to tissue atlases.

560 **Methods**

561 **Collection of the SpatialCorpus-110M**

562 **Dissociated data collection**

563 We collected and combined dissociated single-cell and single-nucleus data from the latest patch of
564 CellXGene¹⁰¹, 50 additional curated studies available through the sfaira data zoo⁵⁷, 150 datasets acquired
565 through the Gene Expression Omnibus (GEO) data repository^{55,56} and four datasets from the Human Cell
566 Atlas (HCA) data explorer¹⁰².

567 For the data originating from CellXGene, we used the CZ CellXGene Discover Census¹⁰¹ version 2023-07-15
568 and its Python API to download the latest batch of all data available on the census. The CZ CellXGene
569 Discover Census only contains cells from human or mouse, as well as only gene expression measurements
570 obtained via RNA sequencing. We additionally only downloaded primary data that was marked with the
571 respective identifier in the Census to ensure that cells are not represented multiple times in our collection.
572 Subsequently, we downloaded the entire cell and gene metadata as well as the raw counts and stored them
573 as H5AD on disk.

574 For additional data acquisition, firstly, we selected human and mouse 10x Genomics technology datasets not
575 present in the latest CellXGene patch from the sfaira data zoo, which was acquired and categorized as
576 previously described⁵⁷, and excluded datasets without publicly available raw count matrices. We then
577 downloaded the selected data through the sfaira interface, removed any cells with less than 200 expressed
578 genes, streamlined the feature space of each dataset to Ensembl release 104 (GRCh38) protein-coding
579 genes, applied sfaira metadata streamlining, and applied the Nicheformer metadata scheme. We stored the
580 data for each study from sfaira as individual H5AD objects on disk.

581 Secondly, for the acquisition from the GEO data repository, we focused on GEO IDs previously included in
582 the recent scsimilarity⁴⁴ preprint publication. After cross-checking this list with the other used data sources to
583 avoid duplicated data, we acquired the necessary metadata from the GEO website and the corresponding
584 publications. We downloaded the count matrices, converted the various data formats into AnnData format
585 and combined them with the collected metadata to save them as individual H5AD objects on disk. We
586 curated ontology term identifiers for species based on the ontology representation of the NCBI organismal
587 taxonomy (NCBITaxon)¹⁰³, tissue based on the Uber-anatomy ontology (Uberon)^{104,105}, sex based on the
588 ontology of phenotypic qualities (PATO)^{106,107} and assay based on the Experimental Factor Ontology
589 (EFO)¹⁰⁸. All ontology terms were obtained through the Ontology Lookup Service (OLS)¹⁰⁹.

590 Lastly, we followed the same approach for the four HCA data explorer⁵⁸ datasets as for the GEO datasets.
591 To make the dataset acquisition process reproducible and available to the community, we have shared
592 scripts for downloading and standardizing of all datasets. All data-collection-related code can be found at
593 <https://github.com/theislabs/nicheformer-data>. We additionally implemented a validator to streamline the
594 verification process, ensuring alignment between metadata formats and the data collection schema. A
595 detailed list and overview table of all datasets containing GEO ID, DOI, the number of cells, tissue, assay,
596 and author information can be found in Suppl. Table 3.

597 **Spatial data collection**

598 The spatial part of the SpatialCorpus-110M consists of datasets measured with image-based spatial
599 transcriptomics technologies, namely CosMx, ISS, MERFISH, and 10x Xenium. We collected 60 different
600 datasets across 15 different solid organs. Most of the spatial data collection was collected via the Vizgen
601 data release⁶¹, the 10x Genomics data resource⁶² and the CosMx data resource¹¹⁰. The remaining datasets
602 were collected through the data resources stated in the original publications. Unpublished datasets were
603 obtained prior to publication via the original authors. Each dataset was downloaded and stored as individual
604 H5AD files. For each dataset, we collected expression data and associated gene and cell-level metadata, but
605 high-resolution images and segmentation masks were not collected and curated. We curated ontology term
606 identifiers for species based on the ontology representation of the NCBI organismal taxonomy
607 (NCBITaxon)¹⁰³, tissue based on the Uber-anatomy ontology (Uberon)^{104,105}, sex based on the ontology of

phenotypic qualities (PATO)^{106,107} and assay based on the Experimental Factor Ontology (EFO)¹⁰⁸. All ontology terms were obtained through the Ontology Lookup Service (OLS)¹⁰⁹. For Xenium and CosMx assays official ontology terms are not yet defined, so we replaced them with placeholders. For datasets that did not provide Ensembl gene identifiers, we used pyEnsembl⁶³ with the Ensembl release 104 (GRCh38) to map gene names to Ensembl gene identifiers and subsequently BioMart⁶⁴ through the official Ensembl releases⁶⁵ for mapping mouse genes to orthologous gene identifiers. Scripts for acquiring the spatial data are also shared in our GitHub repository. We used the same validator as used for the dissociated datasets to streamline the verification process of the collected metadata. We applied no additional quality control, gene or cell level filtering for the spatial omics datasets beyond the filters applied by the original authors of the publications or the filters automatically applied by the individual spatial transcriptomics technologies. A detailed list and overview table containing the GEO ID, DOI, the number of cells, tissue, assay, and author information for the spatial datasets can be found in Suppl. Table 4.

Datasets used for downstream tasks and evaluations

Publically available datasets used for downstream tasks and evaluations were collected in the same way as the other spatial transcriptomics datasets present in the SpatialCorpus-110M. As most of our downstream tasks require cell type, niche and region label annotations, we focussed primarily on annotated and large-scale spatial transcriptomics datasets. We provide a detailed description of those datasets below.

MERFISH mouse brain

Yao et al.¹⁰ measured 4.3 million cells across 59 tissue sections from one whole male mouse brain using multiplexed error-robust fluorescence in situ hybridization (MERFISH) with a 500-gene panel. This dataset contains a hierarchical cell type annotation structured into four nested levels of annotation. We used the *class_label* field with 33 distinct cell types as input for the Nicheformer niche regression task (Suppl. Fig. 3C), the *division_id* label, containing seven distinct labels (CBX-MOB-other neuronal, Immune, Low Quality - LQ, Neuroglial, PAL-sAMY-TH-HY-MB-HB neuronal, Pallium glutamatergic, Subpallium GABAergic, Vascular), as niche labels (Suppl. Fig. 3B), and the *clean_region_label* field, containing 17 distinct labels (CB, CTXsp, HB, HIP, HY, Isocortex, LSX, MB, OLF, PAL, RHP, STRd, STRv, TH, sAMY, ventricle, white_matter), as the region label (Suppl. Fig. 3A) for the Nicheformer label prediction tasks. The tissue niches represent the cellular organization in the brain, grouping together neurons by major brain structure (pallium, subpallium, hypothalamus/extended amygdala, thalamus/epiphysis, and mid/hindbrain) as well as major neurotransmitter type (glutamate and GABA)¹⁰. Non-neuronal cells are grouped into neuroglial, immune, and vascular niches. The train-test split defined for this dataset is composed of a random image or tissue section hold-out across all sections in the measured entire male mouse brain (Suppl. Fig. 3A-C).

CosMx human liver

We collected the CosMx human liver dataset from the publicly available CosMx data resource¹¹⁰. The dataset comprises cells from both a normal healthy liver measuring 332,877 cells across 301 fields of views covering one tissue section in a male 35-year-old patient, as well as cells from a Hepatocellular Carcinoma measuring 460,441 cells across 383 fields of view in one tissue section from a 65-year-old female patient. Both samples were measured with the 1000-plex CosMx Human Universal Cell Characterization Panel. The dataset includes both cell type and niche labels. For the niche label prediction task, we used the healthy liver section, which provides six distinct labels defining structural zones in the liver: Portal vein (Zone 1a), Zone 1b, Zone 2a, Zone 2b, Zone 3a, and Central vein (Zone 3b) (Suppl. Fig. 5B, D). We did not use the cancer liver sample for the niche label prediction task as it was primarily composed of cells annotated as a general tumor niche without further substructures provided. For the niche composition prediction task, we used both the cancer and healthy liver sections with the cell type labels, which define 22 distinct cell types (Antibody secreting B cells, CD3+ alpha beta T cells, Central venous LSECs, Cholangiocytes, Erythroid cells, Hep, Hep 1, Hep 3, Hep 4, Hep 5, Hep 6, Inflammatory macrophages, Mature B cells, NK like cells, Non inflammatory macrophages, Periportal LSECs, Portal endothelial cells, Stellate cells, gamma delta T cells 1, tumor 1, tumor 2 and an undefined type (NotDet) (Suppl. Fig. 5E). The train-test split defined for this dataset is composed of a random field of view hold-out across both tissue sections (Suppl. Fig. 5A, D).

657 CosMx human lung

658 We collected the CosMx human lung dataset from the publicly available CosMx data resource¹¹⁰. This
 659 dataset contains samples from five different donors (301,611, 89,975, 227,110, 71,304 and 81,236 cells,
 660 respectively) across eight fields of views measured with the 1000-plex CosMx Human Universal Cell
 661 Characterization Panel. All donors have just one field of view, with the exception of the first donor, which has
 662 three fields of view and the third, which has two fields of view. The train-test split defined for this dataset is
 663 composed of a random field of view hold-out (Suppl. Fig. 7A-B). CosMx provides both cell type and niche
 664 labels. We use the 22 distinct cell type labels defined in this dataset for the niche composition prediction
 665 task. These labels are B-cell, NK, T CD4 memory, T CD4 naive, T CD8 memory, T CD8 naive, Treg,
 666 endothelial, epithelial, fibroblast, mDC, macrophage, mast, monocyte, neutrophil, pDC, plasmablast, tumor
 667 12, tumor 13, tumor 5, tumor 6, tumor 9 (Suppl. Fig. 7C).

668 Xenium human lung

669 We collected the Xenium human lung dataset from the 10x Genomics data resource
 670 (<https://www.10xgenomics.com/datasets>). This dataset measures two different lung sections, an adult human
 671 healthy lung (295,883 cells) and an adult human lung with invasive adenocarcinoma (531,165 cells). Both
 672 sections are measured with the 289-plex Xenium Human Lung Gene Expression Panel and an additional
 673 100 lung-cell type-specific genes. As this dataset is not annotated, we only use it for the neighborhood
 674 density prediction task. We computed a spatial graph of cells with a radius of 25 μm^2 to calculate the cellular
 675 niche densities. The train-test split defined for this dataset is a random cell hold-out across all cells from both
 676 sections.

677 Xenium human colon

678 We collected the Xenium human colon dataset from the 10x Genomics data resource
 679 (<https://www.10xgenomics.com/datasets>). This dataset measures two different colon FFPE-preserved tissue
 680 sections: a non-diseased colon (275,822 cells) and a cancer stage 2A adenocarcinoma (587,115 cells). Both
 681 sections are measured with the 325-plex Xenium Human Colon Gene Expression Panel and an additional
 682 100 genes specifically selected to cover signaling and chemokine genes, and markers for stromal cells. As
 683 again this dataset is not annotated, we only use it for the neighborhood density prediction task. We
 684 computed a spatial graph of cells with a radius of 17 μm^2 in both sections to calculate the cellular niche
 685 densities. The train-test split defined for this dataset is a random cell hold-out across all cells from both
 686 sections.

687 Dissociated dataset used for label transfer

688 Single-cell RNA-seq of the primary motor cortex

689 Yao et al. generated a large-scale transcriptomic and epigenetic atlas of the mouse primary motor cortex¹¹.
 690 We subsetting this large-scale dataset to cells measured with 10x v3 single-cell RNA seq. The subset
 691 captures 21,884 genes in 7,416 cells and annotates 19 different cell types (Astro, Endo, L5 ET, L5 IT, L6 CT,
 692 L6 IT, L6 IT Car3, L6b, L2/3 IT, L5/6 NP, Lamp5, Microglia, OPC, Oligo, Pvalb, Sncg, Sst, CLMC and Vip)
 693 (Fig. 3C). We manually transferred cell types present in this dataset to the cell types measured in the
 694 MERFISH mouse brain dataset. Respectively, we mapped Astro to Astro-Epen; Endo and VLMC to Vascular;
 695 Microglia to Immune; Oligo and OPC to Oligo; L6 IT, L6 IT Car3, L5 IT, L2/3 IT, L5 ET to IT-ET Glut; L5/6 NP,
 696 L6b and L6 CT to NP-CT-L6b Glut; and Lamp5, Sncg, Vip Pvalb and Sst to CGE/MGE GABA.

697 Nicheformer tokenization, architecture and pretraining

698 Nicheformer tokenization

699 The Nicheformer training corpus encompasses over 110 million cells in total, measured in more than 350
 700 datasets using 8 different sequencing technologies and two species: human and mouse. The total number of
 701 genes considered is 20,310, being 16,981 orthologous genes, 3,178 human-specific and 151
 702 mouse-specific. For Nicheformer, we use a tokenization strategy similar to the one in Geneformer²¹ with the

difference that the cell transcripts are normalized according to the technology-specific non-zero mean to account for differences in the sequencing protocol. First, all cells are normalized so that each of them has 10,000 counts. To account for technological variations, we then compute a technology-specific gene expression non-zero mean vector i.e., the mean expression value of each gene, without considering the zero counts. We computed a single dissociated mean expression vector for the dissociated datasets because the differences between sequencing protocols in the dissociated cells are not as large as in the spatial assays. We then normalize the expression of each cell using the corresponding technology-specific mean expression vector to obtain the expression of each gene in each cell relative to the whole training corpus. Finally, the genes are ranked in descending order, from most expressed to lowest expressed, excluding all not expressed genes, creating an ordered set T of genes:

$$T = \left\{ \text{idx}(gex_0), \text{idx}(gex_1), \dots, \text{idx}(gex_n) : gex_{norm_i} \geq gex_{norm_{i+1}} ; gex_{norm_i} \neq 0 \right\}$$

where $\text{idx}(gex_i)$ is a function that returns the index of gene i in a previously defined vocabulary of genes and gex_i is the gene expression of gene i of a cell. To incorporate the influence of biological context on gene expression, we prepend contextual tokens for <ASSAY>, <MODALITY> and <ORGANISM> to the set T to incorporate metadata information to the input data. These tokens encode metadata information, such as assay type (e.g., MERFISH, CosMx, 10x 5' v2, etc.), modality (dissociated or spatial), and organism (mouse or human). Recognizing the significant impact biological context can have on gene expression, we augment the input sequences for our transformer model with modality, organism, and assay tokens. This approach allows the model to explicitly learn representations that account for context-driven variations, leading to more robust and generalizable downstream analyses. Therefore, for a cell i , with a specific assay, organism and modality, the ordered set of tokens T^i is the following:

$$T^i = \left\{ \text{assay}^i, \text{organism}^i, \text{modality}^i, \text{idx}(gex_0^i), \text{idx}(gex_1^i), \dots, \text{idx}(gex_n^i) \right\}$$

As a last step, the length of the set T^i is truncated to $N = 1500$. As not all cells have the same number of expressed genes, there might be sets whose total length is lower than 1500. In those cases, <PAD> tokens are appended such that the final length is $N = 1500$. <PAD> tokens ensure that all inputs have the same length by filling empty spaces with no semantic meaning. This is an important element when handling cells belonging to both RNA-seq and spatial assays because gene panels are usually smaller in the latter, which leads to a larger amount of <PAD> tokens in the set.

735 Nicheformer architecture

Given an initial input set $x^i \in R^{N \times D}$ composed of N tokens of dimensionality D , Nicheformer encodes the position within the set by adding positional embeddings. Instead of modeling as sinusoidal embeddings, we use learnable embeddings for each position¹⁶.

Nicheformer is composed of 12 stacked transformer blocks such that the output of one block is in the input of the following block. Given an input sequence $x^i \in R^{N \times D}$:

$$x_0^i = x^i$$

$$x_{l+1}^i = \text{transformer_block}_l(x_l^i) \quad \forall l \in [0, n - 1]$$

Each transformer block consists of two main modules: a multihead self-attention mechanism and a feed-forward neural network. The multi-head self-attention mechanism enables the model to weigh the relevance of different input elements in the input set when generating output representations²⁵. In our case, we use 16 attention heads, token dimensionality $D=512$ and dimensionality of the hidden layer of the FFN of

1024. The <PAD> tokens are masked for the attention mechanism so that no token can pay attention to them.

Nicheformer pretraining and performance optimization

Nicheformer optimizes masked language modeling (MLM) loss¹⁶ during pretraining. We mask 15% of the tokens, including contextual and gene tokens but excluding <PAD> tokens, during pretraining. The model is then trained to predict the original tokens that have been masked, utilizing the unmasked tokens as context. Specifically, following the BERT schema¹⁶, if the i -th token is chosen to be masked, 80% of the time it is replaced by a <MASK> token, 10% of the time by another random gene or contextual token and 10% of the time it remains unchanged. Mathematically, the MLM loss is described as:

$$L_{MLM} = E_{x \sim X} E_M \sum_{i \in M} \left[-\log p(x_i | x_{[1, n] \setminus M}) \right]$$

where M is the set of masked tokens, X is the entire dataset, x is a cell of the dataset and x_i is gene i of the cell x .

Nicheformer was pretrained for approximately 10 days using three compute nodes, each with four Nvidia A100 40GB GPUs (total 12 GPUs). We train the model using bfloat16 mixed precision. We use the AdamW optimizer¹¹¹ with $\beta_1 = 0.9$ and $\beta_2 = 0.999$, weight decay of 0.1, and dropout of 0.0. The batch size is nine and the gradients are accumulated during 10 batches before running the backward pass. The minimum learning rate is 1e-5, which increases until 1e-3 with a linear warmup of 100,000 steps. After the warmup, a cosine decay regime¹¹² is applied. Gradient clipping is set to 1.0 during the first epoch and then decreased to 0.5. All weights are initialized using Xavier initialization¹¹³ with default parameters, while the bias terms are initialized to 0. Checkpoints were taken every 10000 steps.

Downstream tasks

Spatial cell type, niche and region label prediction

For the spatial cell type, niche and region label classification task, we use the respective labels defined in the individual datasets (see section on Datasets used for downstream tasks and evaluations). We extracted the unique labels for each class, transferred them to 64-bit signed integer values and one-hot encoded them as a matrix with n different classes, with n being either the number of cell types, niches or region. We then used for linear probing a linear layer optimized with a cross-entropy loss. We trained on the training set of the respective dataset for one epoch at a learning rate of 1e-3 and with a batch size of 256. The performance metrics reported are calculated on a held-out test set. We selected the model-assigned class label by calculating the argmax over the output vector of the linear layer. Classification uncertainties reported in this work are the output of the linear layer rescaled to $[0, 1]$ such that the sum equals 1 using a Softmax function.

Neighborhood composition

For the neighborhood composition regression tasks, we first define a spatial graph of cells by building an adjacency matrix based on the Euclidean distance in the two-dimensional coordinate space provided by the individual datasets. The adjacency matrix of spatial cells is a block-diagonal matrix $A \in R^{n \times n}$, with n the number of cells present in the dataset calculated based on the spatial proximity of cells where connectivities can only occur within a field of view. We use a binary adjacency matrix with $a_{ij} = 1$ if $d(x_i, x_j) \leq \delta_r$ where $d(\cdot, \cdot)$ describes the Euclidean distance between nodes $i, j \in n$ and δ_r the is the maximal distance between cells, and $a_{ij} = 0$ otherwise. We do not include self-connectivities for the adjacency matrix to not confound the signal. We additionally define the matrix of observed cell types $X_l \in \{0, 1\}^{n \times l}$ as a one-hot encoding of the l distinct cell types present in the dataset. The neighborhood composition for a given radius is then given as

$$N_r = \text{softmax}(A \times X_r) \in [0, 1]^{n \times l}.$$

The resulting matrix reflects for each cell captured in the dataset a vector giving the proportions of cell types present in the neighborhood of the cell. For the neighborhood prediction task we used a for linear probing a linear layer followed by a Softmax function to rescale the prediction to lie in the range [0,1] and sum to 1. We used the mean square error loss for optimizing this linear layer, trained on the training set of the respective dataset for one epoch at a learning rate of 1e-3 and with a batch size of 256. The performance metrics reported are calculated on a held-out test set.

Neighborhood cell density prediction

For the cellular niche density, we again use the adjacency matrix of spatial cells $A \in R^{n \times n}$ calculated based on the Euclidean distance in the two-dimensional coordinate space. The cellular neighborhood density is then simply given by the row-wise sum of all connectivities in the adjacency matrix.

$$D_r = \sum_j (A_{ij}) \in N^{n \times 1}$$

for all cells present in the dataset with r a given radius, i the index cell for which we want to calculate the density, and j the total number of potential neighboring cells present in the dataset. For the density prediction task, we used for linear probing a linear layer with input being the respective embedding of a cell (Nicheformer, scVI or PCA) and output a scalar. We used the mean square error loss for optimizing this linear layer, trained on the training set of the respective dataset for one epoch at a learning rate of 1e-3 and with a batch size of 256. The performance metrics reported are calculated on a held-out test set.

Nicheformer evaluation, linear probing and fine-tuning

Nicheformer can be fine-tuned or used for linear probing. In both settings, we only train on the previously defined training set of the respective datasets used for downstream tasks (see section on Datasets used for downstream tasks and evaluation). We use in both scenarios all Nicheformer gene tokens extracted from the last layer and average them to get a cell representation. Importantly, the contextual tokens are not used in the aggregation. In linear probing, the previously computed parameter weights of the Nicheformer pretraining model are frozen, i.e., not updated further, and are subsequently used as input to a downstream task. The cell's representation is then fed into a linear layer specific to each downstream task, which represents either a classification task in the case of the niche and region label prediction or a regression for predicting the neighborhood composition and cellular density. For the neighborhood composition task, we additionally fitted a multi-task multilayer perceptron (MLP) that uses the Nicheformer embedding as input and predicts the varying neighborhood composition vectors in a dataset. The MLP is optimized using the average mean squared error across all neighborhood sizes considered. Fine-tuning generally describes using a pretrained model, and training it to a specific downstream task of choice. We speak of a fine-tuned Nicheformer version when we allow the model to change the previously learned parameter space and the weights are updated for a specific task. Importantly, each downstream task can also be optimized with respect to a new set of metrics. All runs are trained for a single epoch with a maximum learning rate of 1e-4 and a cosine decay scheduler reaching 1e-5 at the end. Batch size is nine with gradients accumulated for 10 batches (Suppl. Table 5). We highlight the respective tasks and metrics used to compute them in the section Downstream task.

Baseline comparisons to scVI and PCA embeddings

We compared the performance of the fine-tuned Nicheformer model and the linear probing scenario to embeddings generated with scVI²⁷ and PCA. We split the fine-tuning datasets (MERFISH mouse brain, CosMx human liver, CosMx human lung, Xenium human lung, Xenium human colon) into a training and test set, using the same random splits as applied for the Nicheformer fine-tuning. scVI and PCA were computed on each fine-tuning dataset individually. We used scvi-tools version 1.1.2 with a negative binomial distribution gene likelihood on the raw gene expression counts and trained scVI on the training set with a batch size of

256 for 10 epochs and used 2 hidden layers for the encoder and decoder neural networks. The resulting embedding was chosen to have a latent dimension of 256. After training, we returned the latent representation for each cell in both the training and the test set.

For generating principal component analysis (PCA) embeddings for each dataset we used the implementation available in sklearn version 1.4.1. We first normalized the respective raw gene expression counts for each dataset so that each cell has a total number of counts equal to the median of the total counts for all cells with scanpy version 1.10.1. Next we use scanpy to log1p-transform the data matrix to ensure the data is centered before using it as input to the PCA implementation. We use the sklearn implementation with the number of components being set to 30. All other parameters are the defaults provided by the sklearn implementation. We fit the PCA on the training set and afterwards apply the dimensionality reduction to both the training and test set. The resulting lower dimensional representations, X_{scvi} and X_{pca} then serve as input to a linear layer specific to each downstream task (Suppl. Table 5).

Code and data availability

All models described here are implemented in a Python package available at <https://github.com/theislab/nicheformer>. All public datasets used for pretraining and fine-tuning the models will be made available upon publication as downloading and preprocessing scripts at <https://github.com/theislab/nicheformer-data>.

The Allen brain atlas consortium generated the Allen Institute brain atlas mouse p20, Allen Institute brain atlas mouse p28, and Allen Institute brain atlas mouse female datasets (Suppl. Table 4), which have kindly been provided to us prior to publication. The Xenium human spinal cord, the ISS human brain GBM, the ISS human discover healthy lung, the ISS mouse EAE MS, and the Xenium mouse brain datasets have been generated by Mat Nilsson lab and have been kindly provided to us prior to publication.

Author contributions

F.J.T. conceived the study with the help of A.C.S., and A.T.L.; A.C.S. and A.T.L. contributed equally and have the right to list their name first in their curriculum vitae; F.J.T. supervised the project; A.T.L. and A.C.S. performed the analysis and wrote the code; A.T.L. led the data engineering, model design, implementation and pretraining in discussion with G.P., L.H. and A.C.S. led the data curation efforts for the dissociated data collection; A.C.S. led the data curation efforts for the spatial data collection; M.M. and L.D. supported the data curation efforts for the dissociated data collection; F.D. and R.G. supported the data curation efforts for the spatial data collection; R.G. and L.V. helped to interpret the brain and liver results; A.C.S. designed and created all main figures; A.C.S., A.T.L., G.P., R.G., and F.J.T. wrote the manuscript. All authors read, corrected, and approved the manuscript.

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883 Conflict of interest

884 F.J.T. consults for Immunai Inc., Singularity Bio B.V., CytoReason Ltd, Cellarity, Curie Bio Operations, LLC
885 and has an ownership interest in Dermagnostix GmbH and Cellarity. The other authors declare no conflict of
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887 References

- 888 1. Sikkema, L. *et al.* An integrated cell atlas of the lung in health and disease. *Nat. Med.* **29**, 1563–1577
889 (2023).
- 890 2. Kanemaru, K. *et al.* Spatially resolved multiomics of human cardiac niches. *Nature* **619**, 801–810
891 (2023).
- 892 3. Zhang, P. & Li, S. Human cross-tissue cell atlases: unprecedented resources towards systematic
893 understanding of physiology and diseases. *Signal transduction and targeted therapy* vol. 7 352 (2022).
- 894 4. Domínguez Conde, C. *et al.* Cross-tissue immune cell analysis reveals tissue-specific features in
895 humans. *Science* **376**, eabl5197 (2022).
- 896 5. Williams, C. G., Lee, H. J., Asatsuma, T., Vento-Tormo, R. & Haque, A. An introduction to spatial
897 transcriptomics for biomedical research. *Genome Med.* **14**, 68 (2022).
- 898 6. Du, J. *et al.* Advances in spatial transcriptomics and related data analysis strategies. *J. Transl. Med.* **21**,
899 330 (2023).
- 900 7. Marx, V. Method of the Year: spatially resolved transcriptomics. *Nat. Methods* **18**, 9–14 (2021).
- 901 8. Fischer, D. S., Schaar, A. C. & Theis, F. J. Modeling intercellular communication in tissues using spatial
902 graphs of cells. *Nat. Biotechnol.* (2022) doi:10.1038/s41587-022-01467-z.
- 903 9. Varrone, M., Tavernari, D., Santamaria-Martínez, A., Walsh, L. A. & Ciriello, G. CellCharter reveals
904 spatial cell niches associated with tissue remodeling and cell plasticity. *Nat. Genet.* **56**, 74–84 (2024).
- 905 10. Yao, Z. *et al.* A high-resolution transcriptomic and spatial atlas of cell types in the whole mouse brain.
906 *Nature* **624**, 317–332 (2023).
- 907 11. Yao, Z. *et al.* A transcriptomic and epigenomic cell atlas of the mouse primary motor cortex. *Nature* **598**,
908 103–110 (2021).
- 909 12. He, S. *et al.* High-plex imaging of RNA and proteins at subcellular resolution in fixed tissue by spatial
910 molecular imaging. *Nat. Biotechnol.* **40**, 1794–1806 (2022).
- 911 13. Lu, Y. *et al.* Spatial transcriptome profiling by MERFISH reveals fetal liver hematopoietic stem cell niche
912 architecture. *Cell Discov* **7**, 47 (2021).

- 913 14. Bommasani, R. *et al.* On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.*
914 *07258* (2021).
- 915 15. Brown, T. B. *et al.* Language Models are Few-Shot Learners. *arXiv [cs.CL]* (2020)
916 doi:10.48550/ARXIV.2005.14165.
- 917 16. Devlin, J., Chang, M.-W., Lee, K. & Toutanova, K. BERT: Pre-training of Deep Bidirectional Transformers
918 for Language Understanding. *arXiv [cs.CL]* (2018).
- 919 17. Dosovitskiy, A. *et al.* An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale.
920 *arXiv [cs.CV]* (2020) doi:10.48550/ARXIV.2010.11929.
- 921 18. Gemini Team *et al.* Gemini: A Family of Highly Capable Multimodal Models. *arXiv [cs.CL]* (2023).
- 922 19. Ji, Y., Zhou, Z., Liu, H. & Davuluri, R. V. DNABERT: pre-trained Bidirectional Encoder Representations
923 from Transformers model for DNA-language in genome. *Bioinformatics* **37**, 2112–2120 (2021).
- 924 20. Yang, F. *et al.* scBERT as a large-scale pretrained deep language model for cell type annotation of
925 single-cell RNA-seq data. *Nature Machine Intelligence* **4**, 852–866 (2022).
- 926 21. Theodoris, C. V. *et al.* Transfer learning enables predictions in network biology. *Nature* **618**, 616–624
927 (2023).
- 928 22. Rives, A. *et al.* Biological structure and function emerge from scaling unsupervised learning to 250
929 million protein sequences. *Proc. Natl. Acad. Sci. U. S. A.* **118**, (2021).
- 930 23. Lin, Z. *et al.* Evolutionary-scale prediction of atomic-level protein structure with a language model.
931 *Science* **379**, 1123–1130 (2023).
- 932 24. Madani, A. *et al.* Large language models generate functional protein sequences across diverse families.
933 *Nat. Biotechnol.* **41**, 1099–1106 (2023).
- 934 25. Vaswani, A. *et al.* Attention Is All You Need. *arXiv [cs.CL]* (2017).
- 935 26. Ericsson, L., Gouk, H., Loy, C. C. & Hospedales, T. M. Self-Supervised Representation Learning:
936 Introduction, Advances and Challenges. *arXiv [cs.LG]* (2021).
- 937 27. Gayoso, A. *et al.* A Python library for probabilistic analysis of single-cell omics data. *Nat. Biotechnol.* **40**,
938 163–166 (2022).
- 939 28. Virshup, I. *et al.* The scverse project provides a computational ecosystem for single-cell omics data
940 analysis. *Nat. Biotechnol.* (2023) doi:10.1038/s41587-023-01733-8.
- 941 29. Eraslan, G., Simon, L. M., Mircea, M., Mueller, N. S. & Theis, F. J. Single-cell RNA-seq denoising using
942 a deep count autoencoder. *Nat. Commun.* **10**, 390 (2019).
- 943 30. Luecken, M. D. *et al.* Benchmarking atlas-level data integration in single-cell genomics. *Nat. Methods*

- 944 **19**, 41–50 (2022).
- 945 31. Lotfollahi, M. *et al.* Mapping single-cell data to reference atlases by transfer learning. *Nat. Biotechnol.*
946 **40**, 121–130 (2022).
- 947 32. Roohani, Y., Huang, K. & Leskovec, J. Predicting transcriptional outcomes of novel multigene
948 perturbations with GEARS. *Nat. Biotechnol.* (2023) doi:10.1038/s41587-023-01905-6.
- 949 33. Lotfollahi, M., Wolf, F. A. & Theis, F. J. scGen predicts single-cell perturbation responses. *Nat. Methods*
950 **16**, 715–721 (2019).
- 951 34. Lotfollahi, M. *et al.* Predicting cellular responses to complex perturbations in high-throughput screens.
952 *Mol. Syst. Biol.* **19**, e11517 (2023).
- 953 35. Hetzel, L. *et al.* Predicting Cellular Responses to Novel Drug Perturbations at a Single-Cell Resolution.
954 *arXiv [cs.LG]* (2022).
- 955 36. Cui, H., Wang, C., Maan, H., Duan, N. & Wang, B. scFormer: A Universal Representation Learning
956 Approach for Single-Cell Data Using Transformers. *bioRxiv* 2022.11.20.517285 (2022)
957 doi:10.1101/2022.11.20.517285.
- 958 37. Cui, H. *et al.* scGPT: toward building a foundation model for single-cell multi-omics using generative AI.
959 *Nat. Methods* (2024) doi:10.1038/s41592-024-02201-0.
- 960 38. Rosen, Y. *et al.* Universal cell embeddings: A foundation model for cell biology. *bioRxiv* (2023)
961 doi:10.1101/2023.11.28.568918.
- 962 39. Chen, J. *et al.* Transformer for one stop interpretable cell type annotation. *Nat. Commun.* **14**, 223
963 (2023).
- 964 40. Wen, H. *et al.* CellPLM: Pre-training of Cell Language Model Beyond Single Cells. *bioRxiv*
965 2023.10.03.560734 (2023) doi:10.1101/2023.10.03.560734.
- 966 41. Hou, W. & Ji, Z. Assessing GPT-4 for cell type annotation in single-cell RNA-seq analysis. *bioRxiv*
967 (2023) doi:10.1101/2023.04.16.537094.
- 968 42. Hao, M. *et al.* Large Scale Foundation Model on Single-cell Transcriptomics. *bioRxiv* 2023.05.29.542705
969 (2023) doi:10.1101/2023.05.29.542705.
- 970 43. Consens, M. E. *et al.* To Transformers and Beyond: Large Language Models for the Genome. *arXiv*
971 *preprint arXiv:2311.07621* (2023).
- 972 44. Heimberg, G. *et al.* Scalable querying of human cell atlases via a foundational model reveals
973 commonalities across fibrosis-associated macrophages. *bioRxiv* (2023) doi:10.1101/2023.07.18.549537.
- 974 45. He, F. *et al.* Parameter-Efficient Fine-Tuning Enhances Adaptation of Single Cell Large Language Model

- 975 for Cell Type Identification. *bioRxiv* (2024) doi:10.1101/2024.01.27.577455.
- 976 46. Khan, S. A. *et al.* Reusability report: Learning the transcriptional grammar in single-cell RNA-sequencing
977 data using transformers. *Nature Machine Intelligence* **5**, 1437–1446 (2023).
- 978 47. Liu, T., Li, K., Wang, Y., Li, H. & Zhao, H. Evaluating the Utilities of Foundation Models in Single-cell
979 Data Analysis. *bioRxiv* (2024) doi:10.1101/2023.09.08.555192.
- 980 48. Boiarsky, R., Singh, N., Buendia, A., Getz, G. & Sontag, D. A Deep Dive into Single-Cell RNA
981 Sequencing Foundation Models. *bioRxiv* 2023.10.19.563100 (2023) doi:10.1101/2023.10.19.563100.
- 982 49. Kedzierska, K. Z., Crawford, L., Amini, A. P. & Lu, A. X. Assessing the limits of zero-shot foundation
983 models in single-cell biology. *bioRxiv* 2023.10.16.561085 (2023) doi:10.1101/2023.10.16.561085.
- 984 50. Alsabbagh, A. R. *et al.* Foundation Models Meet Imbalanced Single-Cell Data When Learning Cell Type
985 Annotations. *bioRxiv* 2023.10.24.563625 (2023) doi:10.1101/2023.10.24.563625.
- 986 51. Cook, D. P. *et al.* A Comparative Analysis of Imaging-Based Spatial Transcriptomics Platforms. *bioRxiv*
987 2023.12.13.571385 (2023) doi:10.1101/2023.12.13.571385.
- 988 52. Hartman, A. & Satija, R. Comparative analysis of multiplexed in situ gene expression profiling
989 technologies. *bioRxiv* (2024) doi:10.1101/2024.01.11.575135.
- 990 53. Lopez, R. *et al.* A joint model of unpaired data from scRNA-seq and spatial transcriptomics for imputing
991 missing gene expression measurements. *arXiv preprint arXiv:1905.02269* (2019).
- 992 54. Salas, S. M. *et al.* Optimizing Xenium In Situ data utility by quality assessment and best practice
993 analysis workflows. *bioRxiv* 2023.02.13.528102 (2023) doi:10.1101/2023.02.13.528102.
- 994 55. Edgar, R., Domrachev, M. & Lash, A. E. Gene Expression Omnibus: NCBI gene expression and
995 hybridization array data repository. *Nucleic Acids Res.* **30**, 207–210 (2002).
- 996 56. Barrett, T. *et al.* NCBI GEO: archive for functional genomics data sets—update. *Nucleic Acids Res.* **41**,
997 D991–5 (2013).
- 998 57. Fischer, D. S. *et al.* Sfaira accelerates data and model reuse in single cell genomics. *Genome Biol.* **22**,
999 248 (2021).
- 1000 58. HCA Data Explorer. <https://explore.data.humancellatlas.org/>.
- 1001 59. Chen, K. H., Boettiger, A. N., Moffitt, J. R., Wang, S. & Zhuang, X. RNA imaging. Spatially resolved,
1002 highly multiplexed RNA profiling in single cells. *Science* **348**, aaa6090 (2015).
- 1003 60. Ke, R. *et al.* In situ sequencing for RNA analysis in preserved tissue and cells. *Nat. Methods* **10**,
1004 857–860 (2013).
- 1005 61. Data release program. *Vizgen* <https://vizgen.com/data-release-program/> (2021).

- 1006 62. Datasets. *10x Genomics* <https://www.10xgenomics.com/datasets>.
- 1007 63. Perkins, A. & Henze, C. Increasing the efficiency of GEOS-Chem Adjoint model runs using a Python
1008 ensemble manager. (2012).
- 1009 64. Smedley, D. *et al.* BioMart--biological queries made easy. *BMC Genomics* **10**, 1–12 (2009).
- 1010 65. Martin, F. J. *et al.* Ensembl 2023. *Nucleic Acids Res.* **51**, D933–D941 (2023).
- 1011 66. Arendt, D. *et al.* The origin and evolution of cell types. *Nat. Rev. Genet.* **17**, 744–757 (2016).
- 1012 67. Heumos, L. *et al.* Best practices for single-cell analysis across modalities. *Nat. Rev. Genet.* 1–23 (2023).
- 1013 68. Palla, G., Fischer, D. S., Regev, A. & Theis, F. J. Spatial components of molecular tissue biology. *Nat.*
1014 *Biotechnol.* **40**, 308–318 (2022).
- 1015 69. Satija, R., Farrell, J. A., Gennert, D., Schier, A. F. & Regev, A. Spatial reconstruction of single-cell gene
1016 expression data. *Nat. Biotechnol.* **33**, 495–502 (2015).
- 1017 70. Pham, D. *et al.* Robust mapping of spatiotemporal trajectories and cell-cell interactions in healthy and
1018 diseased tissues. *Nat. Commun.* **14**, 7739 (2023).
- 1019 71. Hu, J. *et al.* SpaGCN: Integrating gene expression, spatial location and histology to identify spatial
1020 domains and spatially variable genes by graph convolutional network. *Nat. Methods* **18**, 1342–1351
1021 (2021).
- 1022 72. Wang, Q. *et al.* The Allen Mouse Brain Common Coordinate Framework: A 3D Reference Atlas. *Cell*
1023 **181**, 936–953.e20 (2020).
- 1024 73. Feichtenbeiner, A. *et al.* Critical role of spatial interaction between CD8⁺ and Foxp3⁺ cells in human
1025 gastric cancer: the distance matters. *Cancer Immunol. Immunother.* **63**, 111–119 (2014).
- 1026 74. Barua, S. *et al.* Spatial interaction of tumor cells and regulatory T cells correlates with survival in
1027 non-small cell lung cancer. *Lung Cancer* **117**, 73–79 (2018).
- 1028 75. Galon, J. *et al.* Type, density, and location of immune cells within human colorectal tumors predict
1029 clinical outcome. *Science* **313**, 1960–1964 (2006).
- 1030 76. Fridman, W. H., Pagès, F., Sautès-Fridman, C. & Galon, J. The immune contexture in human tumours:
1031 impact on clinical outcome. *Nat. Rev. Cancer* **12**, 298–306 (2012).
- 1032 77. Fischer, D. S., Schaar, A. C. & Theis, F. J. Learning cell communication from spatial graphs of cells.
1033 *bioRxiv* 2021.07.11.451750 (2021) doi:10.1101/2021.07.11.451750.
- 1034 78. Hildebrandt, F. *et al.* Spatial Transcriptomics to define transcriptional patterns of zonation and structural
1035 components in the mouse liver. *Nat. Commun.* **12**, 7046 (2021).
- 1036 79. Zhang, M. *et al.* Molecularly defined and spatially resolved cell atlas of the whole mouse brain. *Nature*

- 1037 **624**, 343–354 (2023).
- 1038 80. Colonna, M. & Butovsky, O. Microglia Function in the Central Nervous System During Health and
1039 Neurodegeneration. *Annu. Rev. Immunol.* **35**, 441–468 (2017).
- 1040 81. Ben-Moshe, S. & Itzkovitz, S. Spatial heterogeneity in the mammalian liver. *Nat. Rev. Gastroenterol.*
1041 *Hepatol.* **16**, 395–410 (2019).
- 1042 82. Robinson, M. W., Harmon, C. & O’Farrelly, C. Liver immunology and its role in inflammation and
1043 homeostasis. *Cell. Mol. Immunol.* **13**, 267–276 (2016).
- 1044 83. The evolving tumor microenvironment: From cancer initiation to metastatic outgrowth. *Cancer Cell* **41**,
1045 374–403 (2023).
- 1046 84. Parra, E. R. *et al.* Immune cellular patterns of distribution affect outcomes of patients with non-small cell
1047 lung cancer. *Nat. Commun.* **14**, 2364 (2023).
- 1048 85. [No title]. <https://www.10xgenomics.com/datasets/xenium-human-lung-preview-data-1-standard>.
- 1049 86. Armingol, E., Officer, A., Harismendy, O. & Lewis, N. E. Deciphering cell–cell interactions and
1050 communication from gene expression. *Nat. Rev. Genet.* **22**, 71–88 (2020).
- 1051 87. Efremova, M., Vento-Tormo, M., Teichmann, S. A. & Vento-Tormo, R. CellPhoneDB: inferring cell-cell
1052 communication from combined expression of multi-subunit ligand-receptor complexes. *Nat. Protoc.* **15**,
1053 1484–1506 (2020).
- 1054 88. Cang, Z. & Nie, Q. Inferring spatial and signaling relationships between cells from single cell
1055 transcriptomic data. *Nat. Commun.* **11**, 2084 (2020).
- 1056 89. Dimitrov, D. *et al.* LIANA+: an all-in-one cell-cell communication framework. *bioRxiv* 2023.08.19.553863
1057 (2023) doi:10.1101/2023.08.19.553863.
- 1058 90. Nitzan, M., Karaikos, N., Friedman, N. & Rajewsky, N. Gene expression cartography. *Nature* **576**,
1059 132–137 (2019).
- 1060 91. Klein, D. *et al.* Mapping cells through time and space with moscot. *bioRxiv* 2023.05.11.540374 (2023)
1061 doi:10.1101/2023.05.11.540374.
- 1062 92. Haviv, D., Gatie, M., Hadjantonakis, A.-K., Nawy, T. & Pe’er, D. The covariance environment defines
1063 cellular niches for spatial inference. *bioRxiv* (2023) doi:10.1101/2023.04.18.537375.
- 1064 93. Tanevski, J., Flores, R. O. R., Gabor, A., Schapiro, D. & Saez-Rodriguez, J. Explainable multi-view
1065 framework for dissecting inter-cellular signaling from highly multiplexed spatial data. *bioRxiv*
1066 2020.05.08.084145 (2020) doi:10.1101/2020.05.08.084145.
- 1067 94. Vandereyken, K., Sifrim, A., Thienpont, B. & Voet, T. Methods and applications for single-cell and spatial

- 1068 multi-omics. *Nat. Rev. Genet.* **24**, 494–515 (2023).
- 1069 95. Rood, J. E., Maartens, A., Hupalowska, A., Teichmann, S. A. & Regev, A. Impact of the Human Cell
1070 Atlas on medicine. *Nat. Med.* **28**, 2486–2496 (2022).
- 1071 96. Yun, S., Jeong, M., Kim, R., Kang, J. & Kim, H. J. Graph Transformer Networks. in *Advances in Neural*
1072 *Information Processing Systems* (eds. Wallach, H. et al.) vol. 32 (Curran Associates, Inc., 2019).
- 1073 97. Zhao, Q., Ren, W., Li, T., Xu, X. & Liu, H. GraphGPT: Graph Learning with Generative Pre-trained
1074 Transformers. *arXiv preprint arXiv:2401.00529* (2023).
- 1075 98. Liu, J. et al. Towards graph foundation models: A survey and beyond. *arXiv preprint arXiv:2310.11829*
1076 (2023).
- 1077 99. Srivastava, A. et al. Beyond the Imitation Game: Quantifying and extrapolating the capabilities of
1078 language models. *arXiv [cs.CL]* (2022).
- 1079 100. Chiang, W.-L. et al. Chatbot Arena: An Open Platform for Evaluating LLMs by Human Preference. *arXiv*
1080 *[cs.AI]* (2024).
- 1081 101. CZI Single-Cell Biology Program et al. CZ CELL×GENE Discover: A single-cell data platform for
1082 scalable exploration, analysis and modeling of aggregated data. *bioRxiv* 2023.10.30.563174 (2023)
1083 doi:10.1101/2023.10.30.563174.
- 1084 102. Projects - HCA Data Explorer. <https://explore.data.humancellatlas.org/projects>.
- 1085 103. Federhen, S. The NCBI Taxonomy database. *Nucleic Acids Res.* **40**, D136–43 (2012).
- 1086 104. Mungall, C. J., Torniai, C., Gkoutos, G. V., Lewis, S. E. & Haendel, M. A. Uberon, an integrative
1087 multi-species anatomy ontology. *Genome Biol.* **13**, R5 (2012).
- 1088 105. Haendel, M. A. et al. Unification of multi-species vertebrate anatomy ontologies for comparative biology
1089 in Uberon. *J. Biomed. Semantics* **5**, 21 (2014).
- 1090 106. Gkoutos, G. V., Schofield, P. N. & Hoehndorf, R. The anatomy of phenotype ontologies: principles,
1091 properties and applications. *Brief. Bioinform.* **19**, 1008–1021 (2018).
- 1092 107. Gkoutos, G. V., Green, E. C. J., Mallon, A.-M., Hancock, J. M. & Davidson, D. Using ontologies to
1093 describe mouse phenotypes. *Genome Biol.* **6**, R8 (2005).
- 1094 108. Malone, J. et al. Modeling sample variables with an Experimental Factor Ontology. *Bioinformatics* **26**,
1095 1112–1118 (2010).
- 1096 109. Ontology lookup service (OLS). <https://www.ebi.ac.uk/ols4/>.
- 1097 110. He, S. et al. High-plex Multiomic Analysis in FFPE at Subcellular Level by Spatial Molecular Imaging.
1098 *bioRxiv* 2021.11.03.467020 (2022) doi:10.1101/2021.11.03.467020.

- 1099 111. Loshchilov, I. & Hutter, F. Decoupled Weight Decay Regularization. *arXiv [cs.LG]* (2017).
- 1100 112. Loshchilov, I. & Hutter, F. Sgdr: Stochastic gradient descent with warm restarts. *arXiv preprint*
- 1101 *arXiv:1608.03983* (2016).
- 1102 113. Glorot, X. & Bengio, Y. Understanding the difficulty of training deep feedforward neural networks. in
- 1103 *Proceedings of the Thirteenth International Conference on Artificial Intelligence and Statistics* (eds. Teh,
- 1104 Y. W. & Titterton, M.) vol. 9 249–256 (PMLR, Chia Laguna Resort, Sardinia, Italy, 13–15 May 2010).
- 1105